



An adaptive regularized topological segmentation network integrating inter-class relations and occlusion information for vehicle component recognition

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ABSTRACT

In intelligent vehicle damage assessment, component recognition faces challenges such as significant intra-class variability and minimal inter-class differences, which hinder detection, as well as occlusions and ambiguous boundaries, which complicate segmentation. We generalize these problems into three core aspects: inter-object relational modeling, semantic-detail information balancing, and occlusion-aware decoupling. To this end, we propose the Adaptive Regularized Topological Segmentation (ARTSeg) network, comprising three complementary modules: Inter-Class Graph Constraint (ICGC), Constrained Detail Feature Backtracking (CDFB), and Topological Decoupling Segmentation (TDS). Each module is purposefully designed, integrated in a progressive structure, and synergistically reinforces the others to enhance overall performance. Specifically, ICGC clusters intra-class features and establishes implicit topological constraints among categories during feature extraction, enabling the model to better capture inter-class relationships and improve detection representation. Subsequently, CDFB evaluates the impact of channel-wise feature information within each candidate region on segmentation accuracy and computational cost, dynamically selecting appropriate feature resolutions for individual instances while balancing the demands of detection and segmentation tasks. Finally, TDS introduces topological associations between occluded and occluding regions at the feature level and decouples them at the task level, explicitly modeling generalized occlusion regions and enhancing segmentation performance. We quantitatively and qualitatively evaluate ARTSeg on a 59-category vehicle component dataset constructed for insurance damage assessment, achieving notable improvements in addressing the aforementioned problems. Experiments on two public datasets, DSMLR and Carparts, further validate the generalization capability of the proposed method. Results indicate that ARTSeg provides practical guidance for component recognition in intelligent vehicle damage assessment.

1. Introduction

The traditional vehicle insurance process [1] demands significant human and material resources due to the necessity of on-site inspection. The application of intelligent damage assessment in the automobile insurance claims process can substantially streamline key stages, such as accident reporting, vehicle inspection, damage evaluation, and damage verification. By reducing manual intervention and mitigating human error, it enables fast and accurate assessment of vehicle damage. This not

only enhances customer satisfaction with the claims service, but also significantly reduces the claim settlement costs borne by insurance companies [2]. Moreover, it effectively prevents scenarios in which accident vehicles remain on-site awaiting manual inspection, which would otherwise lead to traffic congestion and hinder urban traffic flow [3,4]. At present, the principal emphasis concerning the applicability of intelligent damage assessment in automobile insurance claims is its precision in damage estimation [1,2]. An intelligent assessment system must identify the damaged components, categorize the type of damage, and esti-

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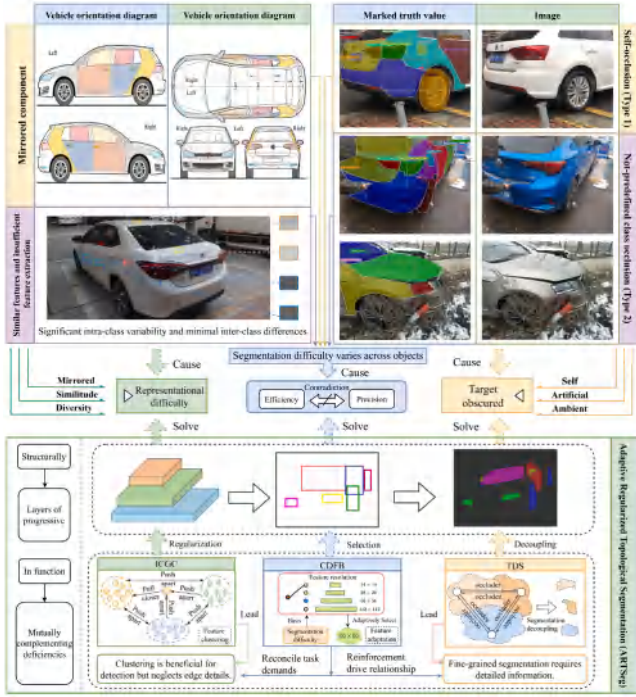


Fig. 1. Current challenges in component detection and segmentation, along with the proposed resolution framework in this paper.

Note: Vehicle mirrored components refer to parts that exhibit mirrored symmetry due to their installation on opposite sides of the vehicle, while remaining identical in geometric structure, appearance, and functionality except for their orientation.

mate its severity. Among these tasks, accurate localization of the components serves as the fundamental prerequisite and critical guarantee for precise damage estimation. To formulate appropriate repair or replacement strategies for the affected component, insurance companies must accurately determine which specific component of the vehicle has sustained damage. Therefore, this study focuses on enhancing the detection and segmentation performance of vehicle components, which is of critical practical relevance and holds considerable potential for real-world applications.

In object detection tasks, significant design variations among vehicles lead to complex intra-class diversity within individual vehicle component categories. Moreover, the symmetrical layout of mirrored components, along with high similarity in textures and colors across different components, results in minimal inter-class distinction. Traditional object detection methods typically process each region of interest independently, making it challenging to accurately differentiate between a wide variety of similar and diverse component types. Additionally, detection performance is a key limiting factor for segmentation quality, as high-precision detection serves as the foundation for fine-grained segmentation [1].

Consequently, optimizing the detection task has become an essential and inevitable requirement.

For segmentation tasks, several challenges arise:

- The customer-provided images of damaged vehicles exhibit significant variability in quality, and the difficulty of segmenting each component's mask also varies accordingly, placing high demands on model robustness. However, existing models typically align features from Regions of Interest (RoI) to a uniform resolution without considering the nature of the target. An excessively low resolution compromises the segmentation of intricate object boundaries, whereas an excessively high resolution results in substantial computational

overhead. Consequently, the varying segmentation difficulties across different targets, coupled with the rigid mask processing strategies employed by conventional models, have led to an intractable trade-off between computational efficiency and segmentation accuracy.

- As an assembly of multiple components, vehicles often exhibit inter-component misalignment and occlusion when imaged on a two-dimensional plane due to varying shooting angles. Additionally, occlusion caused by external objects represents a prevalent phenomenon in real-world scenarios. As a second type of occlusion, it introduces additional complexity to the segmentation task. Because of the complexity and diversity of these external occluders, and the fact that they are not predefined objects to be detected, it is difficult to optimize them using a specific paradigm. Moreover, owing to the vehicle's overall design and aesthetic considerations, the color and texture similarities across components can cause confusion between the contours of occluded components and the boundaries of occluding ones, significantly complicating segmentation in occlusion scenarios.

Therefore, optimizing the segmentation task is both necessary and inevitable.

The aforementioned common problems in vehicle component detection and segmentation can be summarized as follows:

- Due to the similarity in texture and color among components within the same vehicle and the significant variations of corresponding components across different vehicle models, vehicle components as detection targets often exhibit large intra-class variations and small inter-class differences, leading to a high false detection rate;
- Due to the varying segmentation difficulties across different targets in images and the fixed mask processing paradigm adopted by conventional models, there exists an intractable trade-off between computational efficiency and segmentation accuracy;
- Partial occlusion of critical components can occur during the damage assessment process due to non-professional shooting angles or the structural characteristics of the vehicle.

To address the aforementioned three problems, the paper proposes three targeted solutions, as illustrated in Fig. 1.

Therefore, this paper contributes as follows:

- To address problem (i), we propose Inter-Class Graph Constraint (ICGC), a novel approach that constructs class-level representations in the feature space and introduces implicit topological constraints among categories. By modeling the underlying relationships between different categories, ICGC enhances the model's capability to discriminate between visually similar components, thereby improving detection accuracy and feature representation quality.
- To address problem (ii), we propose Constrained Detail Feature Backtracking (CDFB), a learnable mechanism that explicitly models the impact of channel-wise feature information on both segmentation quality and computational cost. By dynamically selecting the most suitable features for each instance, CDFB enables adaptive feature utilization, effectively balancing precision and efficiency in instance segmentation.
- To address problem (iii), we propose Topological Decoupling Segmentation (TDS), which establishes topological associations between occluded and occluding regions in the feature space while decoupling them at the task level. This approach explicitly models occlusion regions in a generalized manner and performs neighbor-level pixel decoupling optimization for all target segmentation boundaries.
- The above three methods are problem-driven in design, progressive in algorithmic structure, and mutually reinforcing in performance, collectively constituting the Adaptive Regularized Topological Segmentation (ARTSeg) network. To validate improvements addressing

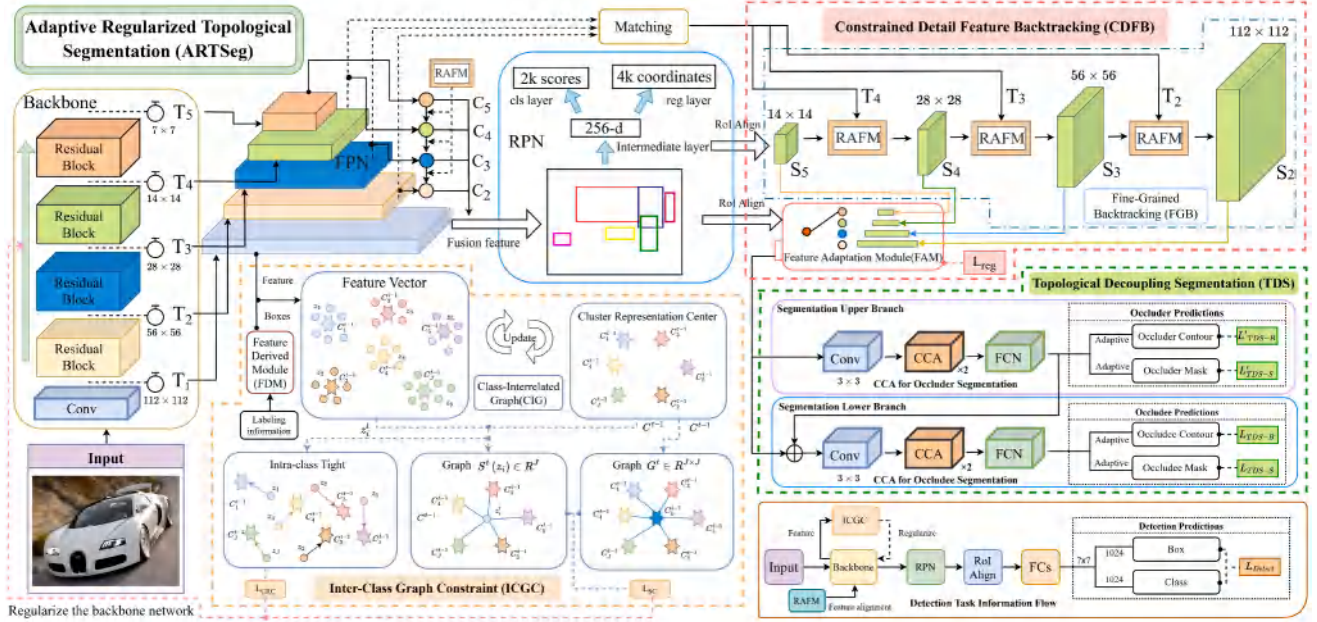


Fig. 2. Overall framework of the adaptive regularized topological segmentation network.

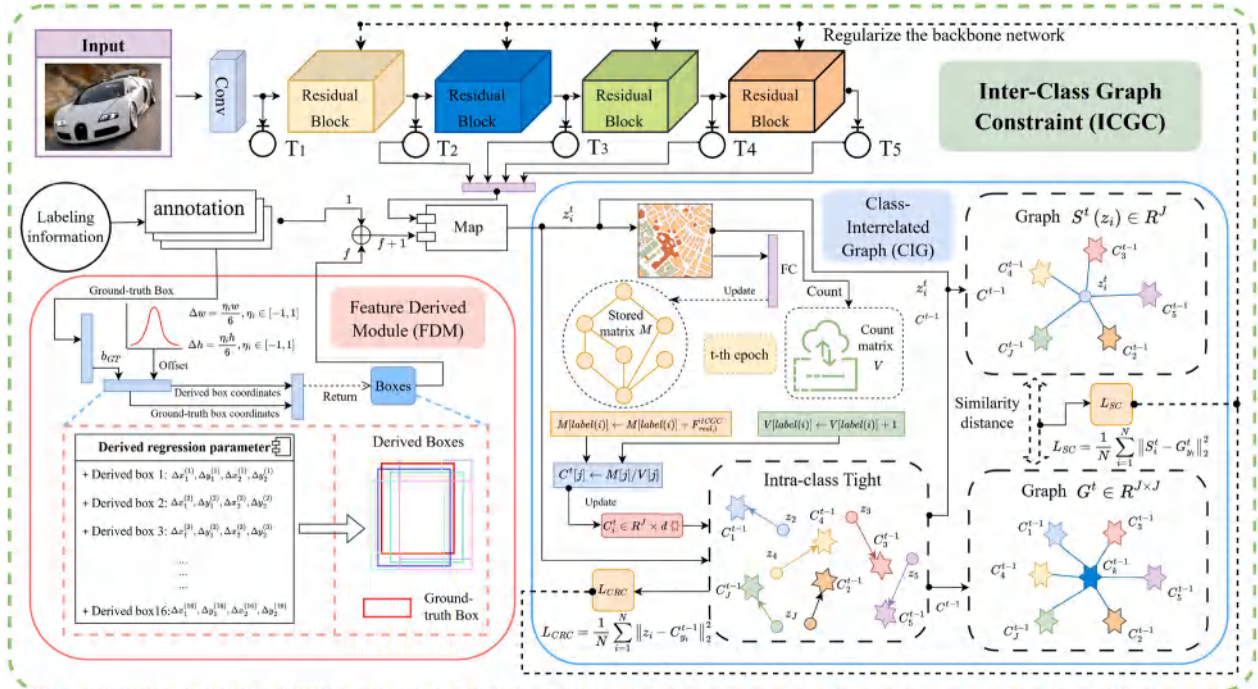


Fig. 3. Inter-class graph constraint.

the aforementioned problems, we quantitatively and qualitatively evaluate ARTSeg on a 59-category vehicle component dataset specifically constructed for insurance damage assessment, and further validate its generalization capability on two public datasets, DSMLR and CarParts.

The remainder of this paper is organized as follows. Section 2 reviews related work on vehicle component recognition, regularized learning, segmentation refinement, and occlusion handling. Section 3 presents the complete architectural framework with detailed descriptions of each constituent module. Section 4 elaborates on the experi-

mental setup, including evaluation metrics, dataset construction, implementation details, and comprehensive validation procedures. Finally, Section 5 concludes the paper and discusses potential future research directions.

2. Related work

2.1. Vehicle component recognition

In application scenarios across various vertical domains, vehicle component recognition tasks are typically implemented using computer

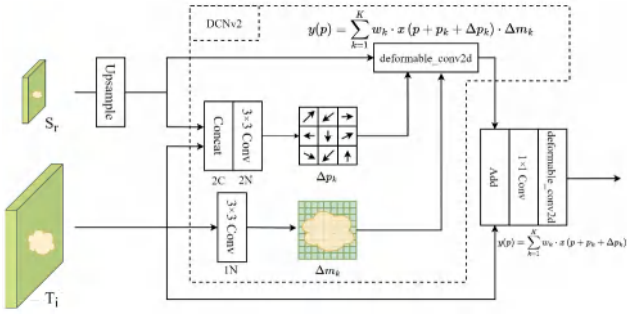


Fig. 4. Region alignment fusion module.

vision techniques such as classification [5,6], semantic segmentation [7,8], object detection [9–11], and instance segmentation [1,2].

However, component recognition in intelligent vehicle damage assessment goes far beyond simply identifying the presence of a door or tire. It requires precise identification of specific vehicle components, differentiation between multiple instances of the same category, and the ability to perform joint analysis with damage information, such as determining which component is damaged and locating the exact area of damage. Therefore, in the context of damage assessment, component recognition constitutes a complex task that is instance-level and multi-label in nature, necessitating the integration of more advanced visual downstream tasks. Simple classification [5,6] or semantic segmentation [7,8] algorithms are insufficient to meet the demands of component recognition in intelligent damage assessment applications.

When object detection algorithms are applied to vehicle component recognition, Xue et al. [9] enhanced the network's ability to detect objects at varying scales through multi-scale training and an improved Non-Maximum Suppression (NMS) algorithm, addressing the issue of overlapping components located in close proximity. Xie et al. [12] proposed a vertically interleaved cross-layer feature fusion strategy to balance the information flow between high-level and low-level nodes, while also transitioning from anchor-based predictions to pixel-based predictions, thereby improving model adaptability across different application scenarios. Liu et al. [10] utilized the YOLO network to detect and localize key vehicle components, estimating vehicle pose by analyzing spatial relationships among the detected components, which aids in vehicle identification and retrieval. Wang et al. [11] proposed a detector based on a diffusion model, which employs latent diffusion to extract multi-level representations that are subsequently fed into the detector head for component detection tasks.

However, intelligent damage assessment requires recognition across all categories of vehicle components. In scenarios involving a large number of component categories, merely using bounding boxes to localize detected components often results in overly dense predictions and excessive redundant regions due to the coarse rectangular boundaries, which falls short of the precision required for real-world applications. To address this limitation, some researchers have explored the integration of pixel-level category differentiation methods (i.e., instance segmentation [13–16]) into component recognition tasks, and have leveraged these techniques to optimize the application of component detection and segmentation.

To facilitate research on vehicle component instance segmentation, the CarParts [17] and DSMLR [18] datasets have been publicly released. Zhai et al. [1] leveraged the characteristics of detection and segmentation tasks to propose an improved decoupled dual-stream network framework for vehicle component detection and segmentation, demonstrating strong generalizability and integrability within existing two-stage instance segmentation frameworks. Zhang et al. [2] implicitly integrate multi-scale semantic knowledge into the model while explicitly incorporating spatial relational information among fine-grained com-

ponents that are otherwise difficult to learn, thereby achieving accurate and generalizable instance segmentation of vehicle components.

These methods significantly improve detection and segmentation performance on conventional vehicle component images. However, existing methods still struggle to achieve satisfactory performance when faced with three challenges commonly observed in intelligent damage assessment scenarios: Problem (i), (ii), and (iii). We will conduct further targeted exploration based on these three specific problem scenarios.

2.2. Regularization problem

The regularization approaches used to assist backbone networks in feature extraction within fine-grained visual classification, along with their demonstrated effectiveness in distinguishing fine-grained features, provide valuable insights for detection and segmentation networks.

Regularization techniques impose constraints or penalty terms during training to uncover hidden topological relationships among data, effectively preventing overfitting and enhancing model generalization. Wang et al. [19] employs Tsallis entropy-based sparsity and small-world structural regularization to constrain the topological structure of feature representations, which effectively enhances the generalization and robustness of the model. A number of prior methods based on metric learning [20,21] have been proposed as regularization techniques for classification tasks, where similarity constraints are imposed to regularize feature encoding and enhance the model's inductive capacity for discriminative feature learning. According to the differences in the form of the metric objectives, metric learning methods can generally be divided into two approaches:

1) One approach is the sample pair-based metric learning paradigm [20,22–24]: it treats data from the same class as positive pairs, minimizing the distance between their embeddings, while treating data from different classes as negative pairs, maximizing the distance between them. Wang et al. [22] first proposed an angular constraint for deep metric learning, which encodes angular relationships among triplets to reveal higher-order relational structures and local information. Zhuang et al. [24] proposed an Attentive Pairwise Interaction architecture to learn mutual features between input sample pairs and incorporate feature prioritization via a score ranking loss. Yuan et al. [20] leveraged intra-image patch-level clustering and localization tasks, using rich feature interactions among image patches to guide more comprehensive feature embedding representations.

2) One approach is the proxy-based metric learning paradigm [21, 25–28]. The introduction of the proxy concept is motivated by the need to optimize metric loss in the face of the combinatorial explosion associated with sample pair formations. Proxy-based metric learning replaces direct distance computation between embeddings with proxy vectors that represent each class. Proxy-NCA [25] was the first to propose a proxy-based metric learning approach. NPT-Loss [26] substitutes the negative proxy in proxy-based triplets with its nearest negative proxy, resulting in a hyperparameter-free formulation and automatically focusing on hard negative mining. HIER [27] captures the latent hierarchical structure of the data by learning hierarchical proxies in hyperbolic space, enabling the sample embeddings to form a tree-like structure aligned with class-level proxies, thereby enhancing the supervisory signal. PSCL-DF [21] leverages prototype learning to extract class prototypes, thereby enhancing the feature discriminability of tail classes.

Conventional instance segmentation models [15,16,29] lack the ability to explicitly model relational constraints across instances. They typically process each RoI independently, without leveraging the relative relationships between object categories, which hinders their ability to distinguish visually similar components. Moreover, traditional metric learning approaches are primarily designed for classification tasks and often fail to generalize effectively to detection and segmentation scenarios. The proposed method falls under the proxy-based metric learning paradigm. However, unlike the aforementioned approaches [20,24,26,27], our Inter-Class Graph Constraint (ICGC) functions as a

self-supervised constraint that captures the latent inter-class correlation among class-level representations. Notably, this differs from traditional image-level similarity metrics. ICGC constructs a regularization term based on instance-level objects represented as proxy vectors, which not only transcends the limitations of classification tasks [21,28] and enables application to downstream tasks such as detection and segmentation, but also enhances the representation of tail categories and models the implicit topological relationships among categories. This, in turn, improves the model's attention to discriminative regions during feature extraction. This component is specifically designed to address Problem (i).

2.3. Segmentation refinement problem

Instance segmentation is a vision task that combines object detection and pixel-level segmentation. It not only requires identifying the category and location of each object but also demands generating precise masks for individual object instances to achieve fine-grained segmentation. This enables the distinction of both different object categories and separate instances within the same category, making it well-suited for complex, multi-object scenarios and aligning closely with the requirements of component recognition in intelligent damage assessment.

Mask-based methods [15,16,29–31] primarily perform pixel-wise classification by determining the instance affiliation of each pixel in a given image. To date, many of the top-performing methods in this category still follow the foundational architecture of Mask R-CNN [13], which typically leverages an object detector to localize each instance within bounding boxes and deploys a refined segmentation head to generate the corresponding object masks. These methods require a uniform feature resolution for all proposed regions during the RoI Align stage to perform predictions. However, low-resolution features are often insufficient to capture the detailed boundary information of hard-to-segment instances, while high-resolution features tend to produce over-segmentation in smooth or homogeneous regions.

To this end, extensive research has been conducted to explore segmentation refinement. Mask Transfuser [14] performs fine-grained, point-wise refinement of coarse boundary masks by constructing a sparse quadtree structure. DCT-Mask [32] encodes high-resolution binary masks into compact vectors via Discrete Cosine Transform (DCT). PatchDCT [30] addresses the issue in DCT-Mask where global vector modifications affect the entire mask, by further refining the DCT vectors at the patch level. In addition to refining the segmentation head, some studies [31,33,34] have proposed coarse-to-fine strategies to enhance mask quality. Cascade Mask R-CNN [34] performs cascading refinement for both detection and segmentation tasks while exploring their interdependencies. PointRend [33] and RefineMask [31] adopt multi-stage architectures to extract fine-grained features. The former performs point-based predictions on ambiguous regions, leveraging adaptive point sampling to refine key areas such as boundaries. The latter enhances boundary quality through multi-stage mask optimization across different feature scales.

However, conventional methods [13–16] typically predict masks with fixed feature resolution, disregarding variations among object instances and failing to account for differences in instance complexity or semantic geometry. While such approaches may suffice for simple cases, they tend to produce overly smoothed predictions on challenging samples, leading to a loss of fine-grained structural details. Inspired by dynamic networks [35,36], which adapt network architectures based on input features, we explore the use of dynamic feature resolutions in instance segmentation to tailor predictions for different instances.

To enhance segmentation performance while maintaining a reasonable computational budget [30,32,35], we introduce the Constrained Detail Feature Backtracking (CDFB) module in this work. CDFB adaptively allocates appropriate feature resolutions based on the segmentation difficulty of individual objects. Specifically, it assigns low-resolution features (with fewer refinement stages) to simple samples to

improve efficiency, and high-resolution features (with more refinement stages) to difficult samples to improve accuracy, thereby achieving a balance between computational cost and segmentation precision. This component is specifically designed to address Problem (ii).

2.4. Segmentation occlusion problem

Occlusion poses a significant challenge in visual perception tasks [37–41]. When multiple objects in an image overlap or partially occlude one another, the model is required to accurately distinguish the boundaries and spatial extents of individual instances under conditions of incomplete visual information. This not only demands strong representational capacity, but also places greater emphasis on the model's ability for spatial reasoning and contextual understanding.

In recent years, various occlusion handling methods have been proposed for different vision tasks [38,40]. OCFusion [37] simultaneously performs Segmentation Completion and Appearance Recovery for occluded vehicles, progressively improving the generation quality through multiple iterative refinements. GIN [39] disentangled external variations via transformation learning and generated invariant shape priors, enabling highly generalized, dynamic generation and precise reconstruction of complete object structures under heavy occlusion. BIFRNet [42] was inspired by the dual visual pathways of the human brain and combined visible features with prior knowledge to restore occluded features, facilitating feature-level occlusion reasoning and robust recognition. Zhang [43] proposed a framework that predicts a global occlusion relation matrix via Holistic Occlusion Relation Inference (HORI) in a single forward pass and performs instance-level generative occlusion completion through a diffusion-based Generative Mask Completion (GMC) module, achieving efficient mutual occlusion handling. Pix2gestalt [44] approached occlusion completion and instance segmentation indirectly by synthesizing complete object images through a diffusion-based image generation framework, rather than directly predicting occluded regions. Yang et al. [45] achieved occlusion-invariant feature learning through pre-task occlusion simulation, a multi-branch shared-weight feature extraction network, and a dynamically balanced joint loss, enabling consistent feature extraction across varying levels of occlusion.

In vehicle imagery, occlusion arises not only from misalignment-induced self-occlusion among the vehicle's own components during imaging, but also from external objects that fall outside the predefined detection targets. The two types of occlusion provide the model with different forms of information that can be exploited. Existing occlusion handling approaches either focus on amodal segmentation [39,44], which attempts to complete occluded regions during inference but requires extensive additional annotations, or are constrained by the ground-truth labels of conventional instance segmentation [37,42], resulting in restricted capability to handle occlusions. In this work, we propose a Topological Decoupling Segmentation (TDS) architecture to address the common challenges posed by both types of occlusions. Specifically, we construct ground-truth annotations for occluded regions by leveraging the relationships between instance masks and bounding boxes. Furthermore, we introduce a parallel occlusion decoupling perception branch within the segmentation head, establishing a unified paradigm for occlusion handling. This design enables the model to explicitly model the interactions between all objects and the target within each RoI during the mask regression phase, and it can be seamlessly integrated into existing two-stage end-to-end instance segmentation frameworks. This component is specifically designed to address Problem (iii).

3. Overall framework and module design

Building upon the problem analysis presented in the Introduction, we propose three targeted solutions. For Problem (i), we introduce a class-level topological constraint method based on intra-class clustering, termed Inter-Class Graph Constraint (ICGC). For Problem (ii), we present a strategy for selecting the feature resolution ratio based on

detail-aware backtracking, referred to as Constrained Detail Feature Backtracking (CDFB). For Problem (iii), we propose a pixel-wise relational modeling approach for handling occlusion, called Topological Decoupling Segmentation (TDS). These three methods, which each tackle different problems, are integrated into a unified framework called Adaptive Regularized Topological Segmentation (ARTSeg).

Within the network framework, ICGC supervises the regularized feature extraction process, CDFB orchestrates feature fusion with a backtracking mechanism to enable adaptive feature selection, and TDS refines the segmentation head to enhance task-specific execution capabilities. Together, these three modules constitute a progressively architectural optimization pipeline, encompassing "regularized representation → feature refinement → execution enhancement".

Algorithm 1: ICGC.

```

Input:
 $T_{pre}^c$ : Warmup epochs;
 $T_{max}$ : Maximum number of epochs;
 $D_{train}$ : Training dataset;
Output:
 $L_{ICGC}$ : Inter-class graph constraint loss;
1 Function Main:
2   Initialize the cluster representation center vectors
    $C^0 \in R^{A \times d}$  with random Gaussian parameters;
3   for  $t \leftarrow 1$  to  $T_{max}$  do
4     Initialize the matrices  $M \in R^{A \times d}$  and  $V \in R^{A \times 1}$  with
       zero vectors;
5     for  $iter = 1$  to  $iters$  do
6       Randomly sample a batch of training data
        $\beta \in D_{train}$ ;
7       Perform bounding box derivation according to
       Eq.(1), generating  $N_r$  bounding boxes and their
       corresponding region features  $f_{real,s}^{(b)}$ ;
8       for  $i \leftarrow 1$  to  $N_r+1$  do
9          $F_{real,i}^{ICGC} \leftarrow MLP(f_{real,s,i}^{(b)})$ : Map the sample
           feature corresponding to the  $i$ -th bounding
           box into the feature space using a fully
           connected layer;
10         $M[label(i)] \leftarrow M[label(i)] + F_{real,i}^{ICGC}$ : Update
           the storage matrix according to the
           corresponding category;
11         $V[label(i)] \leftarrow V[label(i)] + 1$ : Update the
           computation matrix according to the
           corresponding category;
12      end
13      if  $t \geq T_{pre}^c$  then
14        Calculate the cluster representation center
           loss  $L_{CRC}$  using Eq.(17);
15        Calculate the similarity constraint loss  $L_{SC}$ 
           using Eq.(18);
16        Calculate the Inter-class graph constraint loss
            $L_{ICGC}$  using Eq.(19);
17      end
18      Update the model parameters via
       backpropagation;
19    end
20    for  $j \leftarrow 1$  to  $A$  do
21      Update  $C^t[j] \leftarrow M[j]/V[j]$ ;
22    end
23  end
24 return  $L_{ICGC}$ 

```

In terms of model design, the detection and segmentation tasks have divergent preferences in their demands for features. Detection requires

high-dimensional semantic information, while segmentation mainly relies on pixel-level detail information. Therefore, these two tasks exhibit different sensitivities to features at different hierarchical levels. However, we aim to construct inter-class relationships through ICGC to enrich the semantic representations of instance targets and improve detection performance. At the same time, we also aim to build pixel-wise dependency relationships using pixel-level details through TDS to decouple unclear (overlapping) boundaries between targets and enhance segmentation performance. From the perspective of task-specific feature demands, these two goals are contradictory. Although the improvement of detection performance can drive segmentation performance to a certain extent, the detail at segmentation boundaries remains insufficient. To mitigate this contradiction, we propose CDFB to serve as a balancing bridge among semantics and details, objects and pixels, as well as detection and segmentation. Accordingly, ARTSeg utilizes ICGC to regularize the backbone network parameters, shaping high-dimensional semantic features to benefit detection performance. At the same time, it employs CDFB to regress fine-grained features and perform adaptive feature selection, compensating for the loss of detailed information according to the demands of segmentation. This facilitates TDS in modeling pixel-level dependencies and improves segmentation performance. The performance of the three components is clearly mutually reinforcing in design. The overall framework of ARTSeg is shown in Fig. 2.

3.1. Inter-class graph constraint

To address Problem (i), this section introduces Inter-Class Graph Constraint (ICGC), which applies a proxy metric supervision method to construct class-level relationships for regularized training. The goal is to model and characterize object relations exhibiting local Euclidean spatial properties in the feature space. Specifically, ICGC computes class-level representations of samples in the feature space to obtain cluster representation centers that capture the common characteristics of each category, which are then used to construct and update implicit topological constraints among categories. By integrating these constraints with feature learning, the model is guided to represent inter-class relationships more effectively. Through regularizing the backbone parameters, ICGC encourages the network to learn feature representations that are more discriminative between classes, thereby enhancing the model's ability to distinguish between visually similar instances.

The construction and update process of ICGC can be summarized in the following three steps:

- a) The Feature Derived Module (FDM, see Section 3.1.1) derives features based on ground truth annotations to enrich category representations.
- b) The Class-Interrelated Graph (CIG, see Section 3.1.2) generates cluster representation centers online to construct proxy metric vectors.
- c) The similarities between the cluster representation centers, as well as those between the learning samples and each cluster center, are measured and mapped into graph representations. Graph similarity measurement is then performed on both, forming ICGC.

The structure of the ICGC is illustrated in Fig. 3, and the detailed procedure is presented in Algorithm 1.

3.1.1. Feature derived module

The ability of ICGC to model inter-class relationships relies on the assumption that cluster representation centers effectively capture the common features of each category. This, in turn, requires a sufficient number of category proxy samples, as well as the inclusion of additional data variations to enrich intra-class diversity. To address this challenge, we propose a simple yet effective Feature Derived Module (FDM), which enhances the representation in the feature space by synthesizing derived features. These generated training features for selected categories serve to improve data diversity and strengthen the quality of class proxies.

Specifically, the FDM employs a non-learnable bounding box generator to produce region proposals that heavily overlap with the ground-

truth boxes. In contrast to Region Proposal Networks (RPNs), the bounding box generator randomly fine-tunes the ground-truth bounding boxes in the image through coordinate offsets and transforms them into new positive sample proposals. Formally, for a batch of training images I , the designed bounding box generator takes as input the coordinates of each instance in the image, denoted as $b_{GT} = [x_1, y_1, x_2, y_2]$, where (x_1, y_1) and (x_2, y_2) represent the top-left and bottom-right corners, respectively. It then generates $f = 16$ dense derived bounding boxes per instance, whose positions are slightly offset, as follows:

$$\hat{b} = \left[\max(0, x_1 + \frac{\eta_1 w}{6}), \max(0, y_1 + \frac{\eta_2 h}{6}), \min(W, x_2 + \frac{\eta_3 w}{6}), \min(H, y_2 + \frac{\eta_4 h}{6}) \right] \quad (1)$$

Where $\eta_i \in [-1, 1]$ is randomly sampled, and w and h denote the width and height of the bounding box, respectively, with $w = x_2 - x_1$ and $h = y_2 - y_1$. W and H represent the width and height of the image. Each derived box differs slightly from the other 15 in terms of foreground and background composition, which increases sample variance and enhances the representational capacity of the constructed category proxies. By introducing novel derived features, FDM dynamically enhances data diversity, thereby mitigating the issue of insufficiently representative cluster centers discussed in Section 3.1.2.

3.1.2. Class-interrelated graph

Before constructing ICGC, it is essential to establish class-level representations in the form of proxy metric vectors, namely the cluster representation centers. As training progresses, these centers can robustly perform online clustering of category features. Specifically, for J categories, a cluster representation center matrix $C^t \in R^J \times d$ is defined, where each class is represented by a d -dimensional vector. This vector corresponds to the ground-truth region features extracted from instances of the corresponding class, as described in Section 3.1.1, along with the 16 derived region features generated by FDM. The superscript t denotes the t -th epoch during model training. The initial center matrix C^0 is initialized with random Gaussian parameters, such that at the beginning of training, all category centers are random points in the feature space. As training proceeds, the class-level center matrix C^t is iteratively updated at each epoch by averaging all features of the same class.

To measure the target relationships based on the relative distances between cluster representation centers, we preliminarily adopt the cosine similarity function, as it is more effective in capturing similarities in feature dimension distributions. The formulation is as follows:

$$\cos(z_i, z_j) = \frac{z_i \cdot z_j}{\|z_i\|_2 \|z_j\|_2} \quad (2)$$

To ensure the monotonicity and non-negativity of the similarity measurement, the similarity function is further defined as follows:

$$s(z_i, z_j) = \exp(\hat{z}_i \cdot \hat{z}_j / \tau) \quad (3)$$

Where $\hat{z}_i = \frac{z_i}{\|z_i\|_2}$ denotes the normalized feature of z_i , and τ is a temperature parameter that controls the variance.

Given a cluster representation center matrix C^{t-1} used to update the features at the t -th epoch, the j -th row of C , denoted as $C_j \in R^d$, represents the mean feature center of the corresponding category, where $j \in [1, 2, \dots, J]$ and $t \in [1, 2, \dots, T]$. Accordingly, the inter-class relationships can be constructed as follows:

$$G_{kl}^t = s(C_k^{t-1}, C_l^{t-1}) = \exp(\hat{C}_k^{t-1} \cdot \hat{C}_l^{t-1} / \tau) \quad (4)$$

Where G_{kl}^t represents the similarity between the iterative cluster centers C_k^{t-1} and C_l^{t-1} corresponding to any two categories, where C_k^{t-1} denotes the class-level representation obtained using features from the $t-1$ -th epoch. Therefore, for each training epoch, the above computation can be performed across all category pairs, thereby constructing a dynamic class relation graph $G^t \in R^{J \times J}$.

To construct the dynamic topological relationships between a new batch of training samples and the cluster representation center matrix

C^{t-1} , each sample x is first passed through the feature encoder Φ_{fea} to generate its feature representation z . Then, the relative distances between these sample features and the feature centers of all categories are computed and transformed into instance-level graph representations. Specifically, for a sample x_i , its feature extracted at the t -th epoch is denoted as $z_i^t = \Phi_{fea}(x_i^t)$. Based on this, the instance-level graph representation between z_i^t and the cluster representation center C_j^{t-1} from the previous epoch is defined as $S^t(z_i)$:

$$S_j^t(z_i^t) = s(z_i^t, C_j^{t-1}) = \exp(\hat{z}_i^t \cdot \hat{C}_j^{t-1} / \tau) \quad (5)$$

Where $i \in [1, 2, \dots, N]$, $j \in [1, 2, \dots, J]$, and $t \in [1, 2, \dots, T]$. In this way, the instance-level graph representation $S_i^t = [S_1^t(z_i^t), S_2^t(z_i^t), \dots, S_J^t(z_i^t)] \in R^J$ can be generated from the feature of the current sample z_i^t and the dynamic cluster representation center matrix C^{t-1} .

During training, the above steps are used to minimize intra-class variation and generate cluster representation centers, which serve to establish inter-class topological relationship constraints within the model (the corresponding loss formulation is described in Section 3.4.1), thereby promoting the learning of discriminative features. During inference, unseen images are fed into the feature encoder Φ_{fea} , which has been trained under inter-class topological constraints, to extract feature representations that are more conducive to category discrimination.

3.2. Constrained detail feature backtracking

For the segmentation task, vehicle components are tightly packed, and component contours are often difficult to distinguish in the image. Therefore, high-resolution features that contain strong semantic information and precise spatial boundary details are required for salient differentiation.

In Section 3.1, we constructed inter-class relationships through ICGC to enrich and cluster the low-frequency semantic information of category instance targets, aiming to enhance detection performance. However, this regularized clustering strategy, which targets discriminative features, inevitably sacrifices detailed information within object regions, making the model less effective for refining object boundaries in segmentation.

To address Problem (ii), we propose a Constrained Detail Feature Backtracking (CDFB), which consists of the Fine-Grained Backtracking module (FGB) (see Section 3.2.1) and the Feature Adaptation Module (FAM) (see Section 3.2.2). FGB provides fine-grained feature supplementation for regions of interest, while FAM dynamically allocates features at appropriate resolutions for each instance. Together, these components mitigate the loss of detail caused by the regularized feature network, ensuring that the segmentation task's specific feature requirements are met without incurring excessive computational cost. In this way, CDFB selects high-resolution segmentation features that satisfy segmentation demands with moderate computational overhead. The detailed procedure is presented in Algorithm 2.

3.2.1. Fine-grained backtracking

For initial backbone features $\{T_1, T_2, T_3, T_4, T_5\}$, the original Feature Pyramid Network (FPN) [46] introduces a top-down pathway that propagates contextual semantic information from higher to lower layers. In fact, lower-level features contain more fine-grained details than higher-level ones, which is beneficial for high-quality segmentation, especially along object boundaries. However, such information is not fully exploited in two-stage architectures equipped with FPN. To address this, we propose Fine-Grained Backtracking (FGB), which backtracks fine-grained information extracted from shallow layers of the backbone network into the RoI feature layer.

In FGB, we first introduce the Regression Feature Network (RFN), which performs top-down backtracking along the segmentation branch to generate a multi-stage, multi-resolution region-of-interest feature pyramid. Specifically, starting from the RoI-aligned initial region feature

Algorithm 2: CDFB.

Input:
 $f_{RoI}^{(b)}$: Initial RoI Align region features;
 $f_{FPN,i}^{(b)}$: Features extracted from the backbone network
 $\leftarrow \{T_2, T_3, T_4\}$;
State: Model task: *train* and *val*;
 T_{pre}^d : Warmup epochs;
Output:
 $f_{RoI,sel}^k$: Features corresponding to the selected resolution;
 L_{budget} : Budget constraint loss;
 $L_{entropy}$: Information entropy loss;

1 **Function Main:**
2 After RPN and RoI Align with segmentation branch, obtain
 $f_{CDFB,5}^{(s)} = f_{RoI}^{(b)}$;
3 **for** $layer_i : i \leftarrow 4$ **to** 2 **do**
4 $f_{FGB,i}^{(s)} \leftarrow RAFM(f_{FGB,i+1}^{(s)}, f_{FPN,i}^{(b)})$; Perform
 layer-by-layer feature alignment and fine-grained
 backtracking;
5 **end**
6 **if** *State* = *train* **then**
7 **if** $t \leq T_{pre}^d$ **then**
8 Training warm-up for segmentation under
 four-resolution features;
9 **end**
10 **else**
11 $f_{RoI,sel}^k \leftarrow FAM(\{f_{FGB,i}^{(s)}\}_{i=2}^5)$: Predict feature
 resolution using Eq.(8) for subsequent
 segmentation;
12 Calculate the budget constraint loss L_{budget} using
 Eq.(25);
13 Calculate the information entropy loss $L_{entropy}$
 using Eq.(26);
14 Gumbel – Softmax(FAM): Update FAM via
 gradient backpropagation based on Eq.(9);
15 **return** $L_{budget}, L_{entropy}$
16 **end**
17 **end**
18 **else if** *State* = *val* **then**
19 **return** $f_{RoI,sel}^k$
20 **end**

$f_{RoI}^{(b)}$, the network progressively enhances it by backtracking the fine-grained information lost from the initial backbone features $\{T_2, T_3, T_4\}$ during the feature downscaling process, resulting in a top-down regression feature pyramid based on regions of interest, denoted as $\{f_{FGB,5}^{(s)}, f_{FGB,4}^{(s)}, f_{FGB,3}^{(s)}, f_{FGB,2}^{(s)}\}$, and abbreviated as $\{S_5, S_4, S_3, S_2\}$. From S_5 to S_2 , the feature resolution doubles at each stage, reaching 14×14 , 28×28 , 56×56 , and 112×112 , respectively.

Moreover, whether in the feature fusion process of FPN or the backtracking process of FGB, conventional upsampling operations as well as direct feature aggregation methods, such as summation or concatenation, may lead to feature aliasing, thereby reducing the model's performance in detecting and segmenting boundary regions. To address this limitation, we introduce a Region Alignment Fusion Module (RAFM) in FGB to adaptively aggregate the multi-scale backtracked features S with the initial backbone features T .

As shown in Fig. 4, RAFM consists of two deformable convolution layers, which introduce $2K$ learnable offsets to the regular grid sampling positions used in standard convolution. These offsets are learned from the preceding feature maps via an additional convolutional layer. To reduce the incorporation of irrelevant regional features during aggregation, a modulation weight is further introduced to determine whether

a sampled location belongs to a region of interest. If a sampling point falls outside the region of interest, its modulation weight is set to zero. The formulation of the modulated deformable convolution is as follows:

$$y(p) = \sum_{k=1}^K w_k \cdot x(p + p_k + \Delta p_k) \cdot \Delta m_k \quad (6)$$

Where p denotes an arbitrary position on the output feature map and p_k is the k -th sampling location. $\Delta m_k \in [0, 1]$ and $\Delta p_k \in \mathbb{R}$ represent the learnable offset and modulation scalar at the k -th position, respectively, both of which are obtained from a separate 3×3 convolution layer applied to the same input feature map. This convolution layer shares the same spatial resolution as the current convolution layer. It produces an output with $3K$ channels, where the first $2K$ channels correspond to the learned offsets $\{\Delta p_k\}_{k=1}^K$, and the remaining K channels are passed through a sigmoid layer to obtain the modulation scalars $\{\Delta m_k\}_{k=1}^K$.

During the construction of the RAFM module, *deformable_conv2d* from DCNv2 is utilized to adjust the spatial alignment of S_r with T_l , enabling more accurate correspondence while suppressing the incorporation of irrelevant regional features. The *deformable_conv2d* following the 1×1 convolution serves to amplify feature responses in target regions, enhance attention to salient areas, and suppress background interference. Since the feature fusion stage in FPN also involves feature addition, we integrate RAFM into this stage as well. The initial backbone features $\{T_2, T_3, T_4, T_5\}$, after top-down alignment and addition, are denoted in the figure as $\{C_2, C_3, C_4, C_5\}$.

The proposed RAFM is incorporated into both the feature backtracking stage of FGB and the feature fusion stage of FPN to address the issues of feature aliasing and category boundary misalignment during feature aggregation. By aligning high-frequency boundary details between features during both fusion and feature backtracking, RAFM enhances the consistency of intra-class feature representations.

3.2.2. Feature adaptation module

While FGB brings notable performance improvements, it also incurs considerable computational overhead. Prior studies have shown that instances of varying complexity require region features at different resolutions to achieve accurate segmentation. Inspired by this observation, we propose a novel approach to adaptively adjust the resolution of region features for different instances. Instead of relying on the spatial distribution of region features or explicitly identifying regions of interest, our method learns to focus on the impact of channel-wise information in each candidate region on both segmentation accuracy and computational cost. Specifically, we introduce a Feature Adaptation Module (FAM), which predicts the appropriate resolution for backtracked features under constrained computation. This enables an adaptive trade-off between segmentation quality and computational efficiency.

We aim to enable effective cross-channel interaction without dimensionality reduction, thereby learning high-performance and computationally efficient channel attention for adaptive segmentation feature selection. To this end, FAM is designed as a lightweight resolution classifier for feature selection, denoted as $f_{FAM}(\cdot)$, as illustrated in Fig. 5. It consists of a channel attention module, a 1D convolution layer, and a sigmoid activation function. Specifically, given the aggregated features obtained via Global Average Pooling (GAP), FAM generates channel-wise attention weights by performing a fast 1D convolution with a kernel size of T , where T is adaptively determined based on the channel dimension C . The kernel size T defines the extent of cross-channel interaction and is used to model local channel dependencies, enabling layers with a larger number of channels to perform more extensive inter-channel communication. The adaptive kernel size T is computed as follows:

$$T = \psi(C) = \left\lfloor \frac{\log_2(C)}{\gamma} + \frac{b}{\gamma} \right\rfloor_{\text{odd}} \quad (7)$$

Where C denotes the number of input channels, $\gamma = 2$, $b = 1$, and $\lfloor \cdot \rfloor_{\text{odd}}$ indicates rounding to the nearest odd integer.

FAM aims to identify the optimal feature resolution from a set of K candidate feature representations $[r^1, r^2, \dots, r^K]$, enabling subsequent

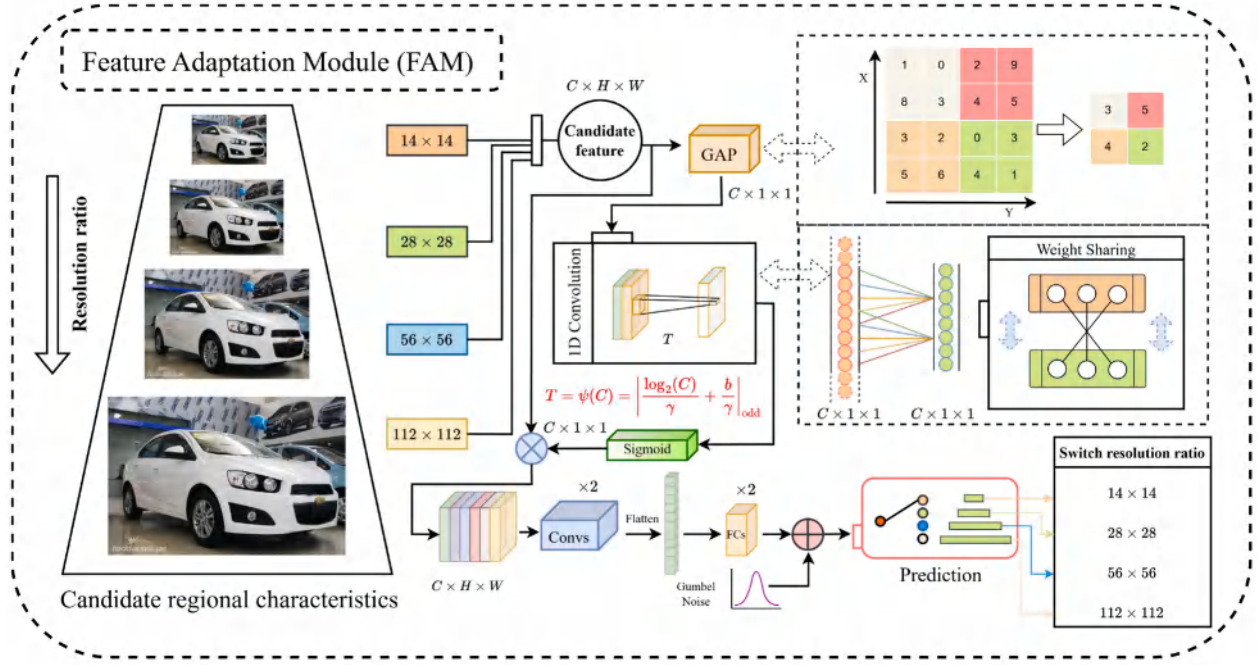


Fig. 5. Feature adaptation module.

segmentation to accurately delineate instances with minimal computational overhead. Specifically, FAM takes the cropped RoI feature as input and produces a probability vector $[p^1, p^2, \dots, p^K]$ via a softmax operation, where each element represents the probability of selecting the corresponding candidate resolution:

$$p^k = \frac{\exp(f_{FAM}^k(x))}{\sum_{t=1}^K \exp(f_{FAM}^t(x))}, k \in \{1, \dots, K\}, t \in \{1, \dots, K\} \quad (8)$$

Where x denotes the RoI feature input to FAM. The candidate resolution with the highest selection probability is selected as the predicted value, which determines the feature resolution size of the target to be segmented.

The soft output P of FAM is transformed into a one-hot prediction, denoted as $Y = [y^1, \dots, y^K]$, where $y^k \in \{0, 1\}$. This transformation can be achieved through discrete sampling; however, such sampling is non-differentiable and thus incompatible with end-to-end training.

To enable gradient backpropagation for updating FAM, we adopt the Gumbel-Softmax reparameterization method. Given a categorical distribution with class probabilities $P = [p^1, p^2, \dots, p^K]$, a set of features can be sampled using the rule $y = \text{one_hot}(\text{argmax}_k(\log p^k + g^k))$, where $\{g^k\}_{k=1}^K$ are independent and identically distributed (i.i.d.) samples drawn from the Gumbel(0, 1) distribution, defined by $g = -\log(-\log(u))$, with $u \sim \text{Uniform}(0, 1)$. We then use the Gumbel-Softmax function as a continuous and differentiable approximation of the original softmax function:

$$y^k = \frac{\exp((\log p^k + g^k)/\tau_0 \cdot \exp(-\alpha t))}{\sum_{t=1}^K \exp((\log p^t + g^t)/\tau_0 \cdot \exp(-\alpha t))} \quad (9)$$

Where $\tau_0 \cdot \exp(-\alpha t)$ denotes the dynamically decaying temperature parameter, which controls the smoothness of the selection process. As $\tau_0 \cdot \exp(-\alpha t)$ approaches zero, the Gumbel-Softmax output becomes increasingly close to one-hot. The initial temperature parameter τ_0 is set to 2, and the decay rate α is set to 0.1. As the training epoch t increases, the temperature parameter decays exponentially, encouraging the model to make more deterministic selections in the later stages of training. The loss computation of FAM is detailed in Section 3.4.3.

3.3. Topological decoupling segmentation

In Section 3.2, we proposed CDFB to select high-resolution segmentation features that meet task requirements with moderate computational overhead. However, in previous two-stage frameworks [30–32], the design of mask heads typically relies on fully convolutional networks to perform binary prediction on occluded targets, ignoring the overlapping relationships between occluders and occluded objects. Furthermore, traditional occlusion handling methods are often limited by ground-truth annotations, and thus are usually limited to addressing occlusions caused by predefined categories.

In view of the above situation, for Problem (iii), we propose a Topological Decoupling Segmentation (TDS) method to solve the common problems of the two types of occlusion. As shown in Fig. 6, TDS relies on the mask boundary relationship to construct the ground truth of the occlusion region. At the same time, it uses the Criss-Cross Attention (CCA) [47] layer to perform pixel topological modeling on the RoI region, obtaining complete global context information and establishing pixel dependency relationships. Additionally, TDS constructs a dual-branch structure for occlusion decoupling perception, explicitly modeling two-dimensional intersection or adjacency relationships between the object boundaries and the mask to handle potential occlusion phenomena, and enabling interaction between the two during regression. TDS can be seamlessly integrated into existing two-stage end-to-end instance segmentation frameworks. The detailed procedure is presented in Algorithm 3.

3.3.1. Pixel-wise topology modeling

Before optimizing the segmentation modeling method, we first need to perform pixel topology modeling to establish dependencies between pixels. Non-local operations [48] directly capture remote dependencies by calculating the interactions between any two locations without being limited to adjacent points. This is equivalent to constructing a convolution kernel of the same size as the feature map, thereby obtaining more detailed information that is well suited to the segmentation task. Additionally, Non-Local operations ensure that the input and output scales are the same, making them easy to integrate as a component with other network structures. However, Non-Local operations are constrained by

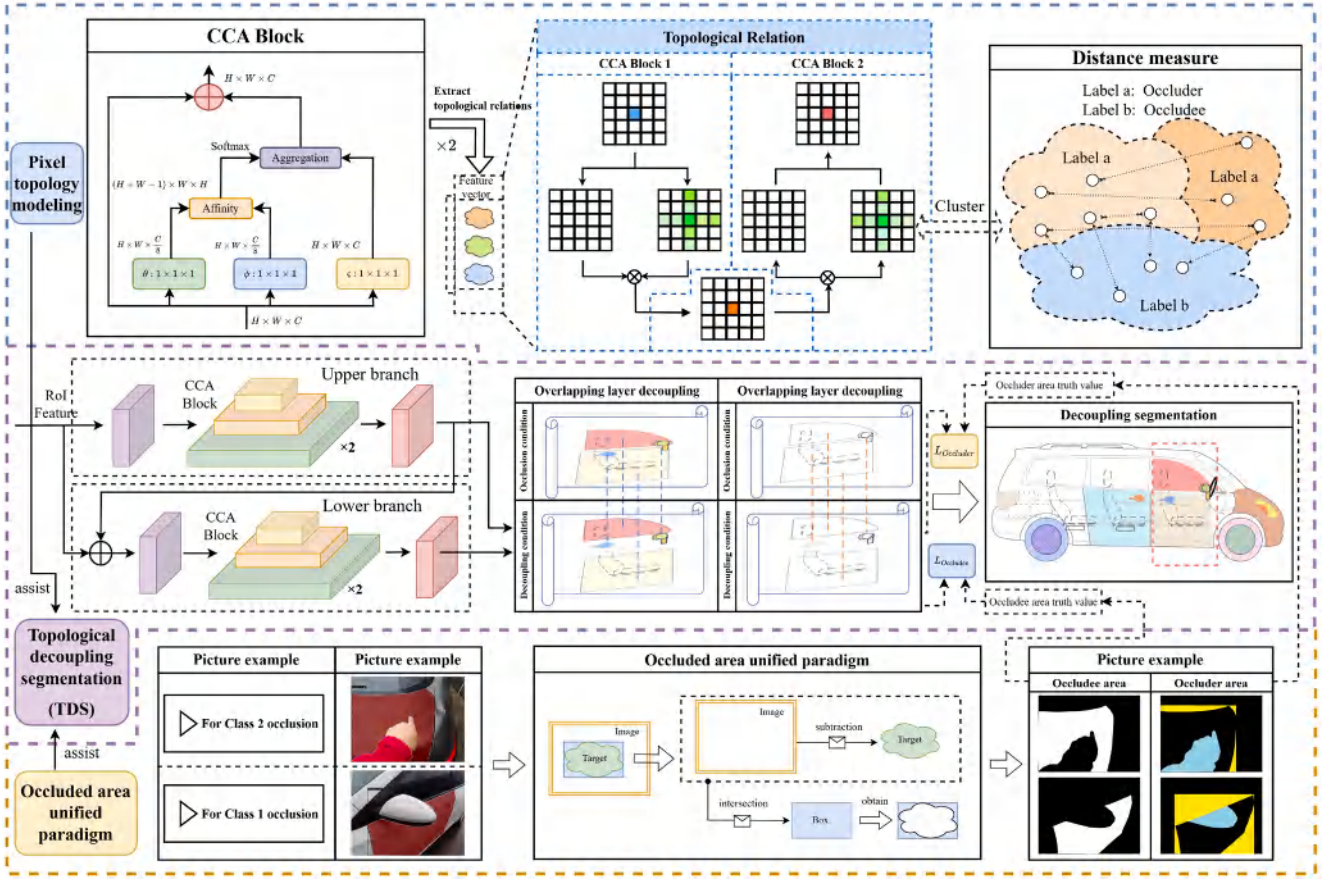


Fig. 6. Topological decoupling segmentation.

their global attention, leading to significant computational complexity $O((H \times W) \times (H \times W))$, which makes them challenging to embed effectively within networks for practical applications.

To address this, we draw inspiration from CCNet [47] and replace Non-local operations with two CCA modules. This substitution allows us to retain dense contextual information and long-range spatial dependencies while significantly reducing both time and space complexity to $O((H + W - 1) \times (H \times W))$. Specifically, given a feature map F obtained after feature resolution selection using FAM (introduced in Section 3.2.2), where $F = f_{\text{Rol, sel}}^k \in R^{C \times W \times H}$. Then, F is processed through three 1×1 convolution modules to generate Q , M , and V , where $V \in R^{C \times W \times H}$, $\{Q, M\} \in R^{C' \times W \times H}$, and C' are set to one-eighth of C to reduce computational complexity. Subsequently, the Q and M undergo an affinity operation to generate an attention map $A \in R^{(H+W-1) \times W \times H}$.

For each position u in Q , a vector $Q_u \in R^{C'}$ can be obtained on the channel dimension. Simultaneously, from M , a set of feature vectors $\Omega_u \in R^{(H+W-1) \times C'}$ corresponding to positions in the same row or column as u is extracted. Let the i -th element of the set Ω_u be denoted as $\Omega_{i,u}$. The affinity is then computed as $d_{i,u} = Q_u \Omega_{i,u}^T$, where $d_{i,u} \in D$ denotes the similarity between Ω_u and $\Omega_{i,u}$, and $D \in R^{(H+W-1) \times W \times H}$ stores all attention weights.

The generated tensor D is normalized using Softmax to obtain the final attention map $A = \text{Softmax}(D)$.

For the generated $V \in R^{C \times W \times H}$, similarly, for each position u , a vector set $\Phi_u \in R^{(H+W-1) \times C}$ can be obtained on the channel dimension. Multiply this vector set by the generated A to perform context information aggregation, as shown in the following formula:

$$F'_u = \sum_{i=0}^{H+W-1} A_{i,u} \Phi_{i,u} + F_u \quad (10)$$

Where F'_u denotes the enhanced feature at position u , and $A_{i,u}$ represents the attention weight coefficient.

Finally, the aggregated contextual information is added back to the local feature map F to produce the output feature F' , thereby enhancing the pixel-level representation capability.

Although the above operations can capture contextual information along both horizontal and vertical directions, a limitation remains: the relationship between position u and any pixel not in the same row or column is still missing. To enable each position to establish connections with all other positions, we apply the criss-cross attention mechanism a second time on F' , resulting in the final output F'' .

The relationship between any two spatial positions can be expressed as:

$$\exists i \in R^{H+W-1}, \text{ s.t. } A_{i,u} = f(A, u_x^{CC}, u_y^{CC}, u_x, u_y) \quad (11)$$

Where $u(x, y) \in R^{H \times W}$ denotes an arbitrary spatial position in the feature map F , and $u^{CC}(u_x^{CC}, u_y^{CC}) \in R^{H+W-1}$ represents an arbitrary position within the criss-cross structure. The function f is a one-to-one mapping that describes the information propagation along the criss-cross attention path.

At this point, all pixels can obtain complete global context information, thereby establishing dependencies between pixels.

3.3.2. Decoupling segmentation

Occlusion can be divided into two types: one is caused by other targets that need to be detected (uniformly referred to as Type 1 occlusion), and the other is caused by objects that are not predefined target categories (uniformly referred to as Type 2 occlusion). Both types of occlusion affect the accuracy of target segmentation. We hope to establish a unified occlusion processing paradigm for both types, which first requires the confirmation of the occlusion region. For Type 1 occlusion,

Algorithm 3: TDS.

Input:
 $f_{RoI,sel}^k$: Features corresponding to the selected resolution;
 N_{total} : Total number of RoI features;
 $State$: Model task: *train* and *val*;
Output:
 $L_{Occluder}$: Segmentation loss for occluders;
 $L_{Occludee}$: Segmentation loss for occluded targets;
 $Mask_{occludee}$: Predicted mask for occluded targets;

1 Function Main:

2 The RoI Align features refined through FGB backtracking are passed through the FAM for resolution selection, where the selected feature $f_{RoI,sel}^k$ is forwarded to the segmentation head;

3 **for** $i \leftarrow 1$ **to** N_{total} **do**

4 $A_{i,u}^{Occluder} \leftarrow CCA(f_{RoI,sel,i}^k)$: The upper branch models pixel-wise dependencies;

5 Calculate the occlusion region feature Z^0 using Eq.(14);

6 Calculate the occlusion-aware feature X_f using Eq.(15);

7 $A_{i,u}^{Occludee} \leftarrow CCA(X_f)$: The lower branch models pixel-wise dependencies;

8 Calculate the occluded target feature Z^1 using Eq.(16);

9 $(Mask_{Occludee}, Contour_{Occludee}) \leftarrow FCN(Z^1)$: Predict the mask and contour of the occluded target;

10 **if** $State = train$ **then**

11 $(Mask_{Occluder}, Contour_{Occluder}) \leftarrow FCN(Z^0)$: Predict the mask and contour of the occluder;

12 Construct the ground truth for occlusion regions based on Eqs.(12) and (13);

13 Calculate the segmentation loss for occluder $L_{Occluder}$ using Eq.(22);

14 Calculate the segmentation loss for occluded target $L_{Occludee}$ using Eq.(23);

15 Update the model parameters using backpropagation;

16 **return** $L_{Occluder}, L_{Occludee}$

17 **end**

18 **else if** $State = val$ **then**

19 **return** $Mask_{occludee}$

20 **end**

21 **end**

the possible occlusion intersection region can be calculated by annotating the category mask and border. However, for Type 2 occlusion, constructing the true occlusion region directly based on the mask border relationship seems difficult because the occluders are not predefined target categories, unlike in Type 1 occlusion.

To establish a unified optimization paradigm for occlusion handling, we take the region corresponding to the ground-truth mask of the currently processed occluded target as the foreground region and the remaining region as the background region, and perform a global inversion operation on the image to obtain the background mask relative to the currently processed target in the image. Ulteriorly, we calculate the intersection region between this background mask and the ground-truth bounding box of the occluded target as the ground-truth region of the target's occlusion. This region includes the above two types of occlusion (as shown by the blue mask region in Fig. 6). It is worth noting that some of the constructed occlusion regions may not represent actual occlusions in the physical space (as shown by the yellow mask region in Fig. 6). However, in the context of 2D images, the adjacent boundaries between these regions and the target to be segmented can still be refined

under a generalized occlusion-optimized paradigm based on the modeled pixel-wise dependencies. At this stage, occlusion regions for both types are unified for construction under a generalized architecture and incorporated into the subsequent occlusion modeling process.

Let the entire image region be denoted as Ω , and the mask region of the current target be M_{obj} . Then, the background mask relative to the current target is defined as:

$$M_b = \Omega \setminus M_{obj} \quad (12)$$

The occlusion region Ω_{occ} is calculated as follows:

$$\Omega_{occ} = \cup (Box \cap M_b) \quad (13)$$

After determining the ground-truth occlusion regions for training, we aim to leverage CCA to capture long-range pixel dependencies, and subsequently decouple the occluder and the occluded target regions. This decoupling helps the model better understand and distinguish occlusion regions and object boundaries. As shown in Fig. 6, we further design the external connections of the CCA in the form of residual connections, constructing an occlusion-decoupling architecture in which the two CCA layers across the upper and lower branches learn the residuals of the input RoI features. Within this architecture, TDS constructs two orthogonal graphs. It is worth noting that the upper branch of TDS consists of a convolutional layer (with a 3×3 kernel), two CCA layers, and an Fully Convolutional Network (FCN) layer. The lower branch adopts a similar architecture, with the primary difference lying in the input and output connection schemes.

The implementation of TDS proceeds as follows: the occluder feature Z^0 output from the upper branch is first transformed by a weight transformation and then added to the original input RoI feature X , resulting in the fused occlusion-aware feature X_f . This fused feature is then used as the input to the lower branch to generate Z^1 , which is employed for predicting the mask of the occluded target. This process effectively associates pixels in the image space based on their similarity, enabling the re-aggregation of features within the target region. As a result, it improves the delineation of object boundaries under occlusion conditions.

In decoupling segmentation architecture, let $f_{RoI,sel}^k \in R^{N \times B}$ denote the input RoI features, where $N = H \times W$ represents the number of pixel grids in the RoI region and B indicates the feature dimension per node (pixel). The complete formulation is given by:

$$Z^0 = \sigma(A^0 f_{RoI,sel}^k W_g^0) + f_{RoI,sel}^k \quad (14)$$

$$X_f = Z^0 W_f^0 + f_{RoI,sel}^k \quad (15)$$

$$Z^1 = \sigma(A^1 X_f W_g^1) + X_f \quad (16)$$

Where W_f denotes the weights of the FCN layer, and $A \in R^{N \times N}$ represents the pixel relation transition matrix defined by Criss-Cross attention (including self-relations). $W_g \in R^{B \times B'}$ is a learnable weight matrix, where $B' = B$ in this implementation. The output feature $Z \in R^{N \times B'}$ consists of node features updated through global information propagation within the entire graph layer. The non-linear function $\sigma(\cdot)$ denotes a normalization operation. The corresponding mask loss and boundary loss constraints will be discussed in detail in Section 3.4.2.

TDS reformulates the traditional class-agnostic segmentation approach into two complementary tasks: occlusion modeling in the upper branch and occluded target prediction in the lower branch. The residual connection between the two branches provides the lower branch with rich occlusion information (such as the shape, location, and boundary of occluded regions) to guide the segmentation of the occluded object. Compared to previous approaches [30,31] where each pixel is assigned a binary foreground/background label within a single-layer mask head, the decoupling segmentation structure establishes a new semantic graph space specifically for occlusion regions. As a result, pixel nodes located in overlapping regions within the RoI can simultaneously correspond to two distinct states in the decoupled graph layer.

In fact, due to the generalized construction of occlusion ground truth, the concept of occluders is extended to include all objects or non-object

regions adjacent to the current target being predicted. When the target to be segmented does not have an actual occlusion relationship (merely shares a common boundary) with adjacent objects in the 2D image, TDS still models this situation as a virtual occlusion pattern between the background (or non-predefined objects) and the target. It then performs boundary refinement accordingly. As a result, the effectiveness of TDS extends beyond handling true occlusion scenarios. It generalizes to decouple and refine the boundaries of all target instances by optimizing the relationships between adjacent pixels, regardless of whether actual occlusion is present.

3.4. Loss function

The proposed framework is trained under the supervision of a composite loss function, which includes inter-class relation graph constraints loss, segmentation loss (comprising both mask and boundary), and penalty loss. The details of each component are elaborated below.

3.4.1. Inter-class graph constraint loss

To better cluster the common features of each category, a non-parametric cluster representation center loss is defined as follows:

$$L_{CRC} = \frac{1}{N} \sum_{i=1}^N \|z_i - C_{y_i}^{t-1}\|_2^2 \quad (17)$$

Where z_i denotes the feature vector of an arbitrary sample, $C_{y_i}^{t-1}$ is the cluster representation center of class y_i generated in the previous epoch (i.e., the $(t-1)$ -th epoch), and $\|\cdot\|_2$ denotes the Euclidean norm.

According to the CIG proposed in Section 3.1.2, the dynamic class relation graph G^t and the instance-level graph representation S_i^t for sample x_i are generated in the t -th epoch. To establish the ICGC based on these relational graphs, we minimize the discrepancy between the dynamic class graph $G_{y_i}^t$ and the instance graph representation S_i^t . This encourages greater intra-class representation similarity while simultaneously enlarging inter-class representation distances. Here, $G_{y_i}^t$ denotes the dynamic class graph representation constructed for sample x_i of class y_i .

Specifically, to measure the distance between graph representations in high-dimensional space while avoiding the computational overhead of square root operations, we adopt the squared Euclidean distance for graph metrics. The similarity constraint loss between the dynamic class relation graph $G_{y_i}^t$ and the instance-level graph representation S_i^t is formulated as:

$$L_{SC} = \frac{1}{N} \sum_{i=1}^N \|S_i^t - G_{y_i}^t\|_2^2 \quad (18)$$

Therefore, the inter-class relation graph constraint loss is defined as:

$$L_{ICGC} = L_{CRC} + L_{SC} \quad (19)$$

ICGC incorporates L_{ICGC} as an additional self-supervised constraint on inter-class relationships, enabling metric-based regularization of the deep representation learning process without introducing additional inference overhead. This constraint effectively captures the implicit topological relationships both within and across classes. By optimizing the structure of feature representations and simplifying the relative positioning of class-level representations, it enhances the model's ability to perceive sample distributions and improves its discriminative power in distinguishing between semantically similar categories.

3.4.2. Segmentation loss

In the TDS architecture proposed in Section 3.3, the cross-validation between features from the upper and lower branches facilitates a more accurate delineation of the true contours of the target to be segmented; however, this process requires appropriate loss functions to constrain and guide the learning.

The mask prediction for the occluder is trained using the mask loss L'_{TDS-S} :

$$L'_{TDS-S} = L_{BCE}(W_S H_{TDS}(f_{RoI,sel}^k), GT_S) \quad (20)$$

Where L_{BCE} denotes the binary cross-entropy loss, H_{TDS} is the non-linear transformation function for occlusion modeling, W_S represents the learnable weights of the mask predictor, $f_{RoI,sel}^k$ refers to the RoI feature refined through fine-grained backtracking and resolution selection, and GT_S denotes the ground-truth occlusion mask computed according to Eqs. (12) and (13).

Numerous studies [14,31,36,40] have empirically demonstrated that mask losses computed from masks of varying quality differ only slightly, making it difficult to reliably distinguish mask quality based solely on mask loss. In contrast, the boundary loss computed from masks at different resolutions exhibits significantly greater variation, offering a more effective signal for assessing mask quality.

The boundary prediction for the occluder is trained using the boundary loss L'_{TDS-B} :

$$L'_{TDS-B} = L_{BCE}(W_B H_{TDS}(f_{RoI,sel}^k), GT_B) \quad (21)$$

Where W_B denotes the learnable weights of the boundary predictor, and GT_B represents the ground-truth boundary of the occlusion region.

At this point, we define the segmentation loss for the occluder, $L_{Occluder}$, and the segmentation loss for the occluded target, $L_{Occldee}$, as follows:

$$L_{Occluder} = \eta_1 L'_{TDS-B} + \eta_2 L'_{TDS-S} \quad (22)$$

$$L_{Occldee} = \eta_3 L_{TDS-B} + \eta_4 L_{TDS-S} \quad (23)$$

Where L_{TDS-B} and L_{TDS-S} denote the boundary loss and mask loss of the occluded target in the lower branch, respectively, analogous to Eqs. (20) and (21), but based on the occluded target feature Z^1 computed in Eq. (16). The hyperparameters $\eta_1, \eta_2, \eta_3, \eta_4$ are employed to balance the contributions of mask and boundary loss components. Based on empirical tuning [40], their values are set to 0.5, 0.25, 0.5, 1.0.

Furthermore, based on the resolution selection mechanism of FAM described in Section 3.2.2, given a positive sample instance x_i , FAM predicts its resolution switching state as $Y = [y^1, \dots, y^K]$. Accordingly, the overall segmentation loss is defined as:

$$L_{segm} = \sum_{i=1}^N \sum_{k=1}^K y^k (L_{Occluder,i}^k + L_{Occldee,i}^k) \quad (24)$$

Where $L_{Occluder,i}^k$ denotes the segmentation loss for the occluder when using the k -th resolution feature of x_i obtained during the backtracking stage. Similarly, $L_{Occldee,i}^k$ represents the segmentation loss for the occluded target. y^k indicates whether the k -th resolution is selected as the output resolution.

3.4.3. Penalty loss

After optimizing the mask and boundary losses as described in Section 3.4.2, the model tends to converge to a suboptimal solution where nearly all instances are predicted using the highest-resolution features (112×112), since these features contain the most detailed information and yield the lowest segmentation loss during training. However, not all instances require such high-resolution features for accurate segmentation. To avoid redundant computation, we introduce a budget-constrained training strategy for FAM, enabling adaptive resolution selection.

Let M denote the computational cost in FLOPs associated with the feature resolution selected for the current batch, and let M_0 represent the target threshold for the FLOPs penalty (defined here as the FLOPs incurred when selecting the 14×14 resolution). When the expected value of M , denoted as $E(M)$, exceeds M_0 , we introduce a progressive soft constraint penalty into the model. This progressive mechanism enables

the model to prioritize the primary task objective, incorporating additional constraints only when necessary. Meanwhile, the use of soft constraints avoids discontinuities in computation, ensuring smoother gradient updates.

In summary, the budget constraint is defined as:

$$L_{budget} = \left[ReLU\left(\sqrt{\frac{E(M)}{M_0}} - 1\right) \right]^p \quad (25)$$

Where p controls the growth rate of the penalty term. When $p = 1$, the penalty increases linearly; when $p = 2$, it increases quadratically. In our setting, we choose $p = 1$.

To prevent the model from overly favoring specific resolutions and to enhance its dynamic adaptability across multiple resolutions, we further introduce an entropy-based loss to balance the resolution prediction of FAM. This encourages the model to fully exploit all candidate resolutions during training.

Given a set of output probability vectors $\{P_1, P_2, \dots, P_N\}$, where N denotes the number of instances in the current batch, the frequency of the k -th feature resolution is computed as $f^k = \frac{1}{N} \sum_i p_i^k$. The entropy loss is then defined as:

$$L_{entropy} = -\frac{1}{K} \sum_k f^k \log f^k \quad (26)$$

When FAM is overly biased toward a specific resolution, the entropy loss increases. Therefore, the entropy loss encourages FAM to maintain a balanced distribution across different feature resolutions, promoting a more uniform selection. This uniformity ensures that the model is effectively trained across all resolutions, preventing insufficient predictive capability for others caused by over-concentration on a single resolution.

L_{budget} directly constrains the computational budget, enforcing the model to maintain efficiency in resource usage. In contrast, $L_{entropy}$ indirectly mitigates the overuse of high-resolution features by encouraging a balanced resolution selection, guiding the model to more uniformly explore the feature selection space across different resolutions. Together, these two components work in synergy to achieve an effective trade-off between computational cost and task performance.

3.4.4. Overall training loss

The entire ARTSeg framework can be trained end-to-end using a multi-task loss function L :

$$L = L_{Detect} + \lambda_1 L_{ICGC} + \lambda_2 L_{segm} + \lambda_3 L_{reg} \quad (27)$$

L_{Detect} is similar to conventional detection loss designs, simultaneously supervising the location prediction and category classification output from the detector. L_{reg} is the regularization term derived from the combination of the budget constraint and entropy loss, i.e., $L_{reg} = 0.5L_{budget} + L_{entropy}$. The coefficients λ_1 , λ_2 , and λ_3 are hyperparameters used to balance each loss function, and are set to 0.0005, 1.0, 0.4, respectively.

4. Experimental validation and analysis

4.1. Experimental preparation

4.1.1. Performance metrics

This study employs mean Average Precision (mAP), a widely adopted evaluation metric in detection and segmentation models, for comprehensive model assessment. The mAP is defined as:

$$mAP = \frac{1}{C} \sum_{n=1}^C \int_0^1 P_n(R_n) dR_n \quad (28)$$

Where C denotes the number of component categories, R_n represents the recall of category n , and $P_n(R_n)$ is the corresponding precision when the recall for category n is R_n . mAP is used to quantitatively evaluate the training performance of the model and reflects its overall effectiveness.

In object detection tasks, mAP is computed based on the Intersection over Union (IoU) between predicted and ground-truth bounding boxes; in segmentation tasks, it is based on the pixel-level IoU between predicted and ground-truth masks.

In this study, Floating Point Operations (FLOPs) and Frames Per Second (FPS) are adopted to evaluate the model complexity. The metric FLOPs refers to the number of floating-point operations required to execute a single forward pass of the network. The metric FPS denotes the number of images the network can process per second.

4.1.2. Dataset and experimental setup

In this study, we first construct a Vehicle Component Dataset (VCD) based on vehicle damage assessment images provided by insurance companies, which is designed for the class-level localization of damaged vehicle components, but is not limited exclusively to damaged components. We annotate the VCD dataset using LabelMe and split it into training and testing sets with a ratio of 8:2, resulting in 19,484 images for training and 4872 images for testing. The dataset covers 59 categories of vehicle components, including components such as the Roof_outer_panel, Outer_mirror (left/right), Tire, Grille, and License_plate. The distribution of different component categories is illustrated in Fig. 7. The images in VCD closely align with real-world intelligent damage assessment scenarios, as they are uploaded during insurance claim processes by different customers. This ensures that VCD faithfully reflects the three major challenges of vehicle component recognition in intelligent damage assessment applications (as discussed in Section 1 and Section 2.1). Subsequent experiments will be conducted on this dataset to validate the effectiveness of the proposed method.

In addition, we further conduct generalization experiments to validate the effectiveness of our method on two publicly available vehicle component datasets: Carparts [17] and DSMLR [18]. The DSMLR dataset includes vehicle types such as sedans, pickup trucks, and SUVs, with annotations for 18 vehicle component categories across 500 images, of which 400 images are used for training and 100 images for testing. The Carparts dataset provides annotations for 23 vehicle component categories, comprising a total of 3833 images, with 3156 images for training, 401 images for validation, and 276 images for testing.

ARTSeg is implemented based on the Detectron2 framework and follows the standard configuration of the baseline Mask R-CNN [13]. Specifically, all experiments are trained for 50 epochs with a batch size of 4, corresponding to a total number of training iterations of 243,550 for VCD, 39,450 for Carparts, and 5000 for DSMLR. The learning rate is decayed to one-tenth of its original value at 85% and 95% of the total number of iterations. The backbone network is initialized with ImageNet-pretrained weights, with BatchNorm layers frozen, while the remaining modules are randomly initialized [14,30]. The warm-up epochs for ICGC and CDFB, denoted as T_{pre}^c and T_{pre}^d , are set to 2 and 4, respectively. During training, we adopt a multi-scale strategy in which the short side of the input image is randomly resized to one of 640, 672, 704, 736, 768, 800, while the long side is constrained to a maximum of 1333 pixels. The hyperparameters of the loss function are set according to empirical tuning heuristics [40], with detailed configurations provided in Section 3.4. During inference, we use the default inference settings. FLOPs are reported as the average number of floating-point operations computed over 100 validation images, and FPS is measured by computing the average runtime per frame with a batch size of 1 on an RTX 3090Ti GPU.

4.2. Experiment

4.2.1. Ablation study

4.2.1.1. Experimental validation of ICGC. To evaluate the proposed ICGC method, along with its constituent components, namely the FDM and the CIG, we conduct ablation studies on the ICGC module based on the baseline, as well as parameter ablation experiments, as shown

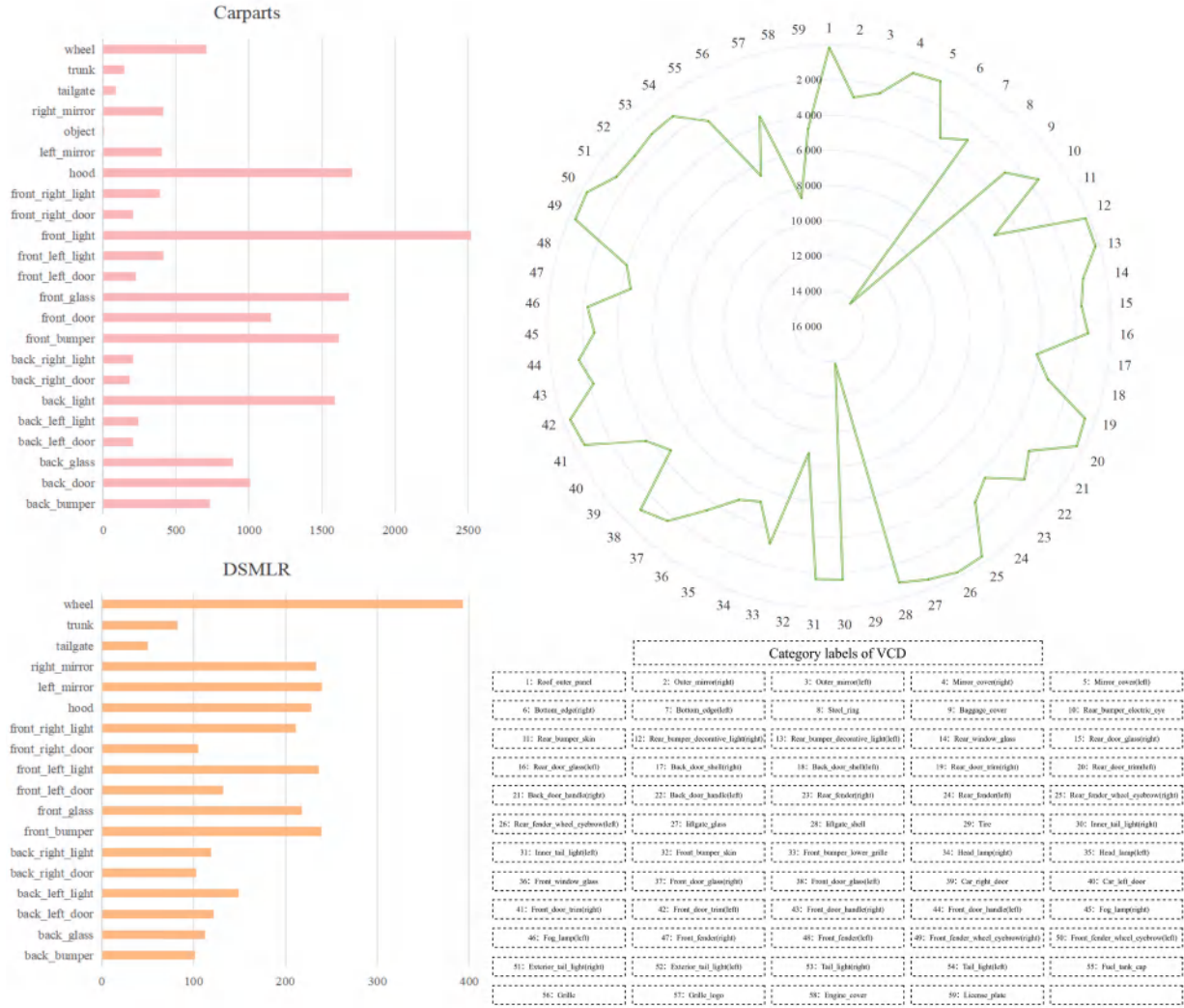


Fig. 7. The category names and category distribution of the datasets VCD, Carparts [17] and DSMLR [18].

Table 1
Ablation study of ICGC on VCD dataset.

Baseline	ICGC		Bbox		Segm		GFLOPs
	CIG	FDM	$mAP_{50}(\%)$	$mAP_{50-95}(\%)$	$mAP_{50}(\%)$	$mAP_{50-95}(\%)$	
Mask R-CNN	✓		59.7	51.3	57.5	44.3	183.4
	✓		64.8	55.9	61.2	47.2	183.4
	✓	$\sqrt{f=4}$	67.0	58.1	63.2	49.1	183.4
	✓	$\sqrt{f=8}$	67.9	58.8	64.1	49.7	183.4
	✓	$\sqrt{f=16}$	68.2	59.0	64.3	49.8	183.4
	✓	$\sqrt{f=24}$	68.0	58.7	63.9	49.5	183.4
	✓	$\sqrt{f=32}$	67.5	58.4	63.3	49.0	183.4

*NOTE: f denotes the number of derived bounding boxes generated by FDM for each instance during training.

in Table 1. Compared to the baseline, incorporating CIG alone enhances the model's ability to capture intra-class common features and to construct implicit topological constraints among categories. This improves the representation of both intra-class and inter-class relationships, thereby enhancing the model's component detection capability. Furthermore, by leveraging the segmentation-driven-by-detection property of the two-stage architecture, the segmentation performance is also improved. On top of this, introducing FDM further increases the sample variance for constructing cluster representation centers in CIG, which alleviates the issue of insufficient representativeness of these centers. This is experimentally reflected in the further improvement of the model's

performance in both detection and segmentation, with the best results observed when the number of derived bounding boxes is set to $f = 16$. However, as f continues to increase (e.g., $f = 24, 32$), a slight performance degradation is observed. Although the model still outperforms the baseline without ICGC, the result indirectly indicates that the FDM exhibits a critical point in its utilization of derived features, beyond which information redundancy and representational saturation begin to emerge. These findings highlight that appropriately setting the number of derived bounding boxes (f) is critical for effective modeling of the cluster structure. Importantly, ICGC regularizes the backbone network parameters during training to encourage learning more discrimi-

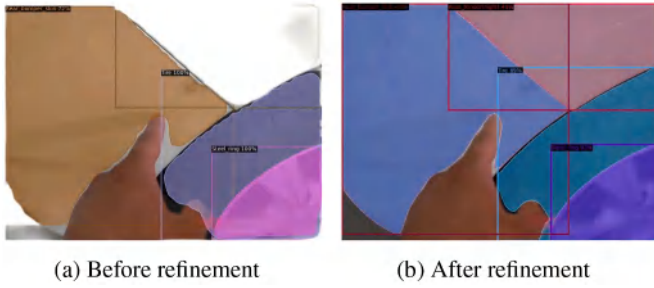


Fig. 8. Necessity of segmentation design.

Table 2

Segmentation performance on VCD dataset under different feature resolutions.

Baseline	Feature resolution	Segm	
		$mAP_{50}(\%)$	$mAP_{50-95}(\%)$
Mask R-CNN	14×14	57.5	44.3
	28×28	58.1	45.4
	56×56	57.8	44.9
	112×112	57.7	44.8

native feature representations, but it does not participate in the inference stage. As a result, it does not introduce additional computational cost or parameter overhead during prediction, as reflected by the unchanged FLOPs metric in the experiment. These experimental results demonstrate that ICGC constructs inter-class relationships by measuring the relative distances between proxy class-level representations. Without increasing inference cost, it enhances the model's ability to learn latent structural dependencies, facilitating a better understanding of the intrinsic properties and salient distinctions in data. This helps the model effectively tackle challenges posed by hidden topological relationships in high-dimensional data, thereby improving its performance and generalization capability when handling complex and similar instances.

4.2.1.2. Necessity of segmentation design. Although ICGC improves detection performance by clustering class-level common features and consequently enhances segmentation performance, it still introduces limitations when handling large-scale objects requiring a broad receptive field. Specifically, the regularized clustering on discriminative features sacrifices fine-grained spatial details in the target regions. After feature clustering and FPN processing, low-level features exhibit enhanced low-frequency semantics, but the detail information is not propagated back to high-level features, resulting in the absence of fine details at the top layers. This hinders accurate segmentation refinement along object boundaries. Although boundary regions occupy only a small proportion of the overall mask and thus have limited influence on the mAP metric, noticeable differences in contour quality are observed in the predicted masks, as shown in Fig. 8. These observations highlight the necessity of further optimization in the segmentation design.

A straightforward approach to improving segmentation quality is to utilize higher-resolution features. However, simply increasing feature resolution often leads to over-segmentation in smooth regions. As shown in Table 2, directly using Mask R-CNN to predict with high-resolution features unexpectedly results in a decrease in mask mAP. This phenomenon can be primarily attributed to the following two reasons:

a). For large-scale objects, their RoI features suffer from intra-class feature inconsistency and feature aliasing due to traditional upsampling and the lack of proper alignment during direct addition or concatenation of features. Consequently, increasing the resolution of these RoI features does not necessarily yield additional useful information for mask prediction.

b). The segmentation head in Mask R-CNN has a relatively simple structure. When the feature grid resolution increases, the segmentation branch lacks sufficient capacity to capture fine-grained features and boundary details, thereby failing to significantly improve prediction accuracy.

The above analysis indicates that merely increasing feature resolution is insufficient to address the problem of segmentation quality. Instead, it is necessary to optimize both the feature alignment and selection strategy (as proposed in our CDFB module) and the design of the mask prediction head (as addressed by our TDS module).

4.2.1.3. Experimental validation of CDFB. Based on the RFN architecture, RAFM is employed for feature alignment in the segmentation RoI feature backtracking stage. As shown in Table 3, under this well-aligned setting, four fixed resolutions (14×14 , 28×28 , 56×56 , and 112×112) and three adaptive strategies (constraint-free, size-based, and FAM) are tested for dynamically selecting feature resolution to evaluate the segmentation performance variations caused by different resolution distributions. Among them, the constraint-free method relies solely on Eq. (24) for accuracy supervision without introducing additional loss to constrain computational cost or resolution assignment probability. In contrast, the size-based method assigns feature resolutions based on object size. Specifically, according to Eq. (29), the four candidate resolutions generated by the RFN backtracking process are allocated to an instance of width w and height h as follows:

$$k = \left\lfloor k_0 + \log_2 \left(\sqrt{\frac{wh}{w_0 h_0}} \right) \right\rfloor \quad (29)$$

Where w_0 and h_0 denote the width and height of the input image, respectively. k_0 represents the index corresponding to the maximum feature resolution (112×112), that is, $k_0 = 4$. Intuitively, Eq. (29) implies that larger objects are assigned higher-resolution features.

Compared with Table 2, which simply increases feature resolution, resulting in performance first increasing and then decreasing, the experiments in Table 3 are based on feature backtracking using the RFN and feature alignment fusion using RAFM for segmented RoI features, achieving segmentation performance that increases with feature resolution. However, this performance gain comes with a substantial increase in computational cost. Furthermore, in Table 3, we compare three adaptive strategies for feature resolution selection. It can be observed that under the constraint-free method, the model tends to converge to a suboptimal solution in which nearly all instances are assigned the highest-resolution features, resulting in considerable computational overhead. The size-based method, on the other hand, allocates resolution solely based on object size, without considering factors such as instance-specific segmentation difficulty or background complexity. This may lead to situations where easily segmentable instances are assigned high-resolution features, while more challenging ones receive lower-resolution features, ultimately limiting performance. In contrast, when FAM adopts 14×14 as the reference resolution for introducing the budget-constrained loss, it achieves a more favorable trade-off between computational cost and segmentation performance compared to the size-based selection strategy. The ablation study on the reference resolution for budget penalty reveals that setting 14×14 as the baseline, where additional budget constraints are introduced only when necessary (i.e., for more challenging segmentation targets), enables efficient use of computational resources. This design allows the model to adaptively select feature resolutions for backtracking based on the complexity of the segmentation task. In contrast, when 28×28 , 56×56 , or 112×112 are used as the reference for penalty, the model increasingly relies on these higher resolutions and tends to overlook the utility of lower-resolution features. As a result, this leads to a substantial increase in FLOPs, while offering only marginal gains in segmentation accuracy.

For the proposed CDFB method, as well as the FGB and FAM involved, we set up baseline ablation experiments, as shown in Table 4.

Table 3

Comparison of segmentation performance on VCD between fixed-resolution features and dynamically selected features.

Feature selector mode		RFN/ RAFM(m)	GFLOPs	Segm		Feature resolution ratio			
				$mAP_{50}(\%)$	$mAP_{50-95}(\%)$	14×14	28×28	56×56	112×112
Fixed resolution	14×14	✓	187.7	57.5	44.3	1	0	0	0
	28×28		202.5	58.4	46.2	0	1	0	0
	56×56		263.1	58.9	46.9	0	0	1	0
	112×112		498.7	59.2	47.4	0	0	0	1
Selective resolution	constraint-free	✓	491.8	59.2	47.2	0.00	0.01	0.01	0.98
	size-based		248.6	58.6	46.5	0.41	0.22	0.24	0.13
	FAM	✓	233.9	59.0	47.2	0.38	0.35	0.18	0.09
			$M_{0,(14 \times 14)}$	59.1	47.3	0.02	0.58	0.24	0.16
			$M_{0,(28 \times 28)}$	59.2	47.3	0.01	0.03	0.73	0.23
			$M_{0,(56 \times 56)}$	59.2	47.4	0.00	0.00	0.01	0.99
			$M_{0,(112 \times 112)}$						

* NOTE: "m" indicates that the RAFM is applied only to the RFN, $M_{0,(14 \times 14)}$ denotes the setting in which the budget loss constraint is introduced based on a reference feature resolution of 14×14 . Similarly, $M_{0,(28 \times 28)}$, $M_{0,(56 \times 56)}$, and $M_{0,(112 \times 112)}$ correspond to resolutions of 28×28 , 56×56 , and 112×112 , respectively.

Table 4

Ablation study of CDFB on VCD dataset.

Baseline	CDFB		Bbox		Segm		GFLOPs	
	FGB		FAM	$mAP_{50}(\%)$	$mAP_{50-95}(\%)$	$mAP_{50}(\%)$		$mAP_{50-95}(\%)$
	RFN	RAFM (b/m)						
Mask R-CNN	$\checkmark$			59.7	51.3	57.5	44.3	183.4
	$\checkmark$			59.6	51.3	57.7	44.8	494.5
	$\checkmark$	$\checkmark$ (b)		66.3	57.2	63.6	48.6	188.5
	$\checkmark$	$\checkmark$ (m)		59.6	51.2	59.2	47.4	498.7
	$\checkmark$	$\checkmark$ (b/m)		66.3	57.1	65.8	50.7	503.9
	$\checkmark$	$\checkmark$ (b/m)	$\checkmark$	66.4	57.2	65.2	50.1	239.1

* NOTE: "✓" indicates that the default feature resolution used for segmentation during the backtracking stage is 112×112 . "b" indicates that RAFM is introduced only into the FPN, "m" indicates that RAFM is introduced only into the RFN, and "b/m" denotes the configuration where RAFM is applied to both FPN and RFN.

Compared to the baseline, introducing the RFN solely in the segmentation branch yields only marginal improvement in segmentation performance at the cost of significantly increased computational overhead. Subsequently, integrating the RAFM into FPN enhances intra-class feature consistency, while incorporating RAFM into RFN enriches boundary details in mask prediction. Both integrations individually improve model segmentation performance. When these two components are combined to construct the complete FGB module, segmentation performance is further improved, as shown in Table 4. However, the deployment of FGB leads to indiscriminate reliance on high-resolution features for segmentation, resulting in excessive computational burden. To address this, FAM is introduced alongside FGB to adaptively assign different resolution ROI features to instances of varying complexity for segmentation. With the inclusion of FAM, the model is able to predict appropriate feature resolutions under controlled computational cost, achieving a balance between segmentation accuracy and efficiency. Experimentally, this results in superior detection and segmentation performance with reasonable computational consumption.

4.2.1.4. Experimental validation of TDS. To evaluate the proposed TDS method, we conduct a set of ablation experiments based on the baseline, with quantitative comparisons presented in Table 5. Starting from the traditional single-branch setting, we first apply a single CCA layer. The results show improved segmentation performance over the baseline, validating the effectiveness of CCA in explicitly modeling pixel-wise dependencies. Next, we incorporate a residual-based occlusion decoupling architecture on top of the baseline. This also results in performance improvement, indicating the positive impact of occlusion modeling under complex scenarios. Further, we evaluate the impact of stacking 1, 2, and 3 CCA layers within the proposed decoupling architecture. As shown in the results, using 2 CCA layers achieves the best segmentation performance with an acceptable computational overhead. Additionally,

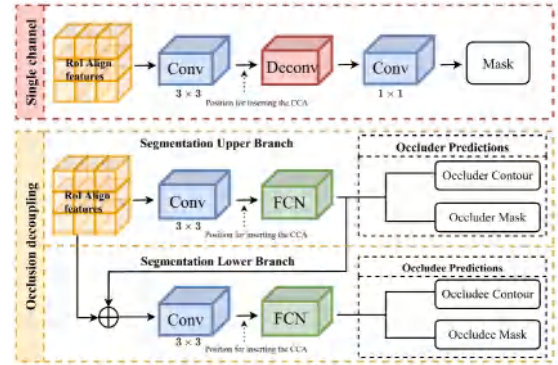


Fig. 9. Structural illustration of the "Single Channel" and "Occlusion Decoupling".

we experiment with replacing CCA with a Non-Local [48] block within the same decoupling architecture. However, despite increasing computational cost compared to the 2-layer CCA configuration, it does not yield better performance. It is worth noting that when using the default feature resolution (14×14), neither increasing the number of CCA layers nor directly using Non-Local introduces significant additional computational overhead when employing TDS independently to enhance the segmentation head (as shown in the upper part of Table 5). However, when TDS is combined with CDFB (m⁺), the increased feature resolution significantly amplifies the computational burden associated with modeling dependencies (see the lower part of Table 5). Therefore, considering the trade-off between performance and efficiency, we adopt a 2-layer CCA configuration for implementing the topological decoupling segmentation strategy.

Table 5
Ablation study of TDS on VCD dataset.

Baseline	CDFB(m ⁺)	Segmentation frame		CCA				Non-Local	Segm		GFLOPs
		Single channel	Occlusion decoupling	0	1	2	3		mAP ₃₀ (%)	mAP ₃₀₋₉₅ (%)	
Mask R-CNN		✓		✓					57.5	44.3	183.4
		✓			✓				58.0	44.9	183.9
			✓	✓					57.9	44.8	188.6
			✓	✓				✓	58.6	46.5	191.9
			✓		✓				58.4	45.9	189.1
			✓			✓			58.9	47.1	189.5
			✓				✓		58.7	46.8	190.0
			✓					✓	59.2	47.5	336.9
✓	✓		✓			✓			59.7	48.4	297.8

* NOTE: To control for variables, a conventional fixed feature resolution of 14×14 is uniformly adopted. The structural illustrations of the "Single Channel" and "Occlusion Decoupling" configurations are shown in Fig. 9. "m⁺" indicates the integration of RAFM into the RFN, with FAM introduced by default.

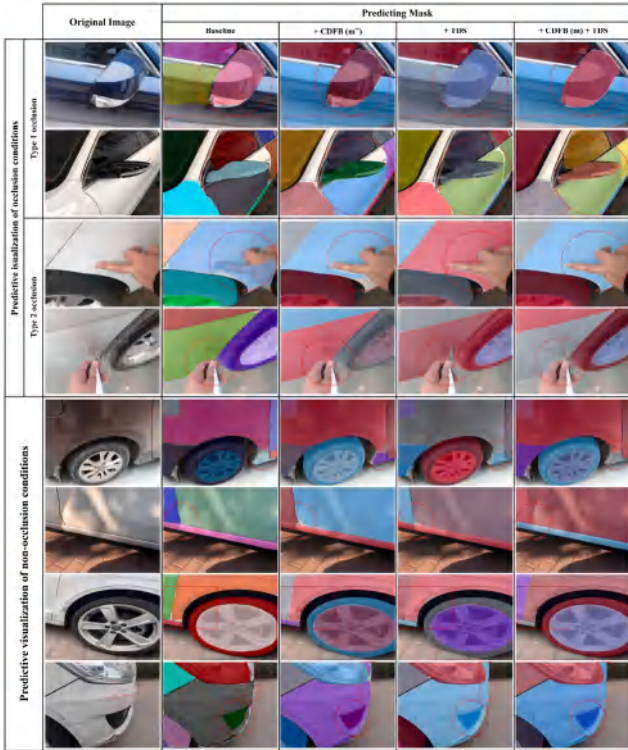


Fig. 10. Visual ablation of individual modules for segmentation performance optimization on VCD dataset.

Note: "m⁺" indicates the integration of RAFM into the RFN, with FAM introduced by default.

4.2.1.5. Effectiveness of segmentation optimization. By introducing the ICGC to regularize the parameters of the backbone network, high-dimensional semantic features at higher layers are effectively shaped to benefit detection performance. Although this indirectly improves segmentation accuracy, it also results in overly coarse mask boundaries. To address this issue, we propose the CDFB module (segmentation optimized by CDFB (m⁺)), which restores fine-grained features and adaptively selects feature resolution based on segmentation requirements to compensate for the loss of detail. Furthermore, we leverage the pixel-wise dependency modeling of the TDS module to perform occlusion-aware optimization during the execution phase of the segmentation. The quantitative performance gains from these segmentation enhancements have been validated in previous experiments. Here, we further present the qualitative segmentation results obtained by separately in-

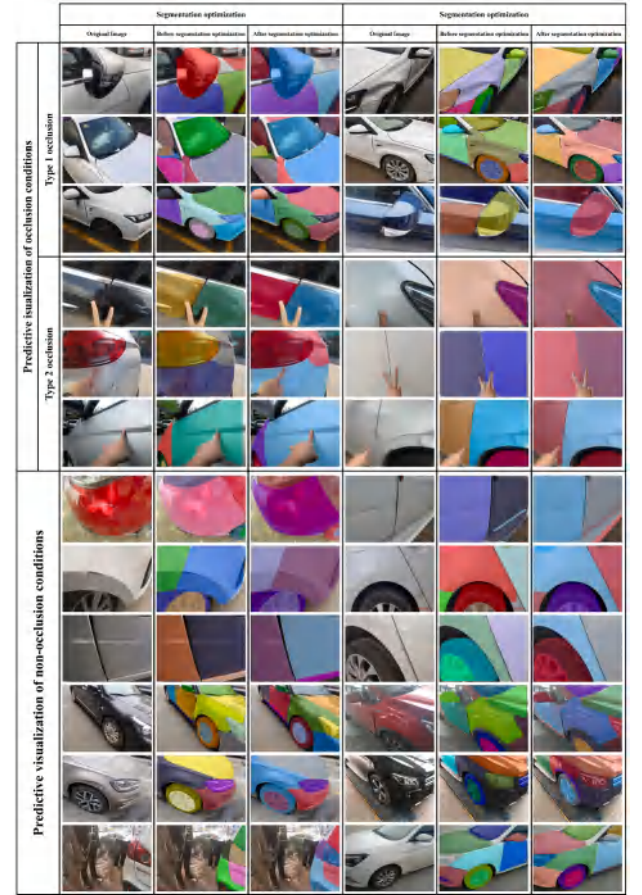


Fig. 11. Comparison of segmentation masks before and after optimization on VCD dataset.

roducing CDFB (m⁺) and TDS, as well as their joint integration. Regions exhibiting significant improvements in boundary segmentation are marked with red boxes or circles for visual emphasis, as shown in Fig. 10. The adaptive feature selection enabled by CDFB (m⁺) and the target-critical boundary refinement introduced by TDS each contribute to improved boundary localization. Differences in the visual outcomes when each module is introduced individually reveal that with only CDFB (m⁺), the model's sensitivity to predefined target regions increases significantly, and pixel-wise classification accuracy for the target improves, yet perception of target boundaries remains problematic. With only TDS, the model's awareness of target-boundary regions is greatly enhanced,

Table 6
Comprehensive module-level ablation study on VCD dataset.

Baseline	Method		Bbox		Segm		GFLOPs	
	ICGC	CDFB	TDS	$mAP_{50}(\%)$	$mAP_{50-95}(\%)$	$mAP_{50}(\%)$		$mAP_{50-95}(\%)$
Mask R-CNN	✓			59.7	51.3	57.5	44.3	183.4
				68.2	59.0	64.3	49.8	183.4
		✓ (m ⁺)		59.6	51.2	59.0	47.2	233.9
		✓ (b)		66.3	57.2	63.6	48.6	188.5
		✓ (b/m ⁺)		66.4	57.2	65.2	50.1	239.1
			✓	59.6	51.4	58.9	47.1	189.5
		✓ (m ⁺)	✓	59.6	51.4	59.7	48.4	297.8
		✓ (b)	✓	66.4	57.1	65.3	50.1	194.6
		✓ (b/m ⁺)	✓	66.4	57.2	66.2	51.5	302.8
	✓		✓	68.1	59.2	66.2	52.5	189.5
	✓	✓ (m ⁺)		68.2	59.1	66.4	52.7	233.9
	✓	✓ (b)		73.2	63.3	70.1	53.4	188.5
	✓	✓ (b/m ⁺)		73.2	63.3	71.8	55.2	239.1
	✓	✓ (b/m ⁺)	✓	73.4	63.4	73.3	56.8	302.8

* NOTE: "b" indicates the integration of RAFM into the FPN; "m⁺" indicates the integration of RAFM into the RFN, with FAM introduced by default; "b/m⁺" represents the integration of RAFM into both FPN and RFN, where FAM is also included by default, corresponding to the full CDFB module configuration.

resulting in boundary delineation that closely matches the true target shape. However, when dealing with complex boundaries, limited feature resolution can lead to mask overspill or shrinkage. When both modules operate together, segmentation performance is optimal. This observation can be explained by the fact that the decoupled modeling optimization in TDS is conditioned on the effective utilization of target features to be segmented. The CDFB (m⁺) module facilitates this process by adaptively selecting features at appropriate resolutions, thereby maximizing feature utility while maintaining computational efficiency. In contrast, if only high-resolution features are supplied to CDFB (m⁺) without a properly optimized segmentation head, the segmentation branch lacks the representational capacity to accurately capture fine boundary structures (an assertion supported by the quantitative results in Table 2). These two modules, therefore, exhibit a complementary relationship and mutually reinforce each other in enhancing boundary perception. Furthermore, Fig. 11 presents a comparison of qualitative segmentation results before and after the overall optimization process. Both Figs. 10 and 11 include representative cases under occluded and non-occluded scenarios.

4.2.1.6. Overall ablation study. To evaluate the effectiveness of the proposed ARTSeg, which integrates the ICGC, CDFB, and TDS, we conduct a comprehensive ablation study involving all components, including separate experiments evaluating CDFB's collaboration with ICGC (focusing on detection performance) and with TDS (focusing on segmentation performance). The quantitative results are summarized in Table 6. Analysis of the results reveals that introducing ICGC substantially improves detection performance and, due to enhanced localization accuracy, also leads to a modest gain in segmentation performance. However, because the regularization process sacrifices some detailed information, the gap between the segmentation and detection metrics widens. In contrast, the introduction of TDS enables a generalized decoupling and relational modeling of boundary pixels associated with the segmentation targets, thereby enhancing the model's capability in boundary discrimination and contour delineation. This leads to improved segmentation performance without affecting detection accuracy. CDFB acts as a bridge between ICGC and TDS. On one hand, CDFB (b) assists in aligning and fusing the features regularized by ICGC, thereby boosting detection performance. On the other hand, CDFB (m⁺) compensates for the loss of detail information according to the demands of segmentation, providing TDS with appropriately-resolved features for pixel-dependency modeling and thus enhancing segmentation performance. Consequently, as shown in Table 6, when all three components are introduced together, both detection and segmentation performance reach their best levels.

4.2.2. Feature visualization experiments

By employing ICGC, ARTSeg regularizes the parameters of the backbone network during training, guiding the model to learn more discriminative feature representations for category separation. Meanwhile, the integration of RAFM into the FPN of the backbone improves intra-class feature consistency. This conclusion is quantitatively supported by the results in Tables 1 and 3. To further validate it, we visualize the features extracted by the network to facilitate a further qualitative comparison at the feature representation level.

First, based on the C_5 feature layer output from the FPN structure in the backbone network, we extract all RoI features corresponding to ground truth instances and perform two-dimensional visualization using t-SNE [50]. For each category, 30 instances are randomly selected for feature visualization, where each point in the figure represents an RoI feature and each color corresponds to a semantic category. The comparison results are shown in Fig. 12. In the baseline model, although most category feature clusters exhibit some clustering tendency, they are not well separated. Some categories clearly display boundary ambiguity or overlapping; for example, left-right mirrored categories such as Exterior_tail_light (left/right) and Front_fender (left/right) show signs of fusion. In contrast, in the visualization of ARTSeg, the cluster boundaries appear significantly clearer and more separable, indicating a stronger discriminative capability across different semantic categories. Moreover, the overall distribution is more structured and dispersed, with larger inter-cluster distances in the t-SNE space, which facilitates subsequent category discrimination. These improvements can be attributed to the ICGC, which serves as an additional supervisory signal during training, enabling the model to incorporate both intra-class feature clustering and inter-class relational encoding into the representation learning process.

We subsequently visualize feature layers $\{C_2, C_3, C_4, C_5\}$ from both ARTSeg and baseline models after backbone feature fusion using Grad-CAM [49], as shown in Fig. 13. Compared to the baseline, ARTSeg demonstrates superior clustering of highly discriminative semantic information in high-level features, consistently observed across both occlusion and non-occlusion scenarios. Furthermore, through FPN's top-down feature fusion mechanism, the model achieves enhanced focus on target regions in shallow features. These advantages collectively improve the discrimination of similar components, thereby boosting detection performance. Notably, while high-level features may lose some details due to clustering regularization, our adaptive backtracking mechanism effectively compensates for segmentation requirements. The superiority of ARTSeg's fea-

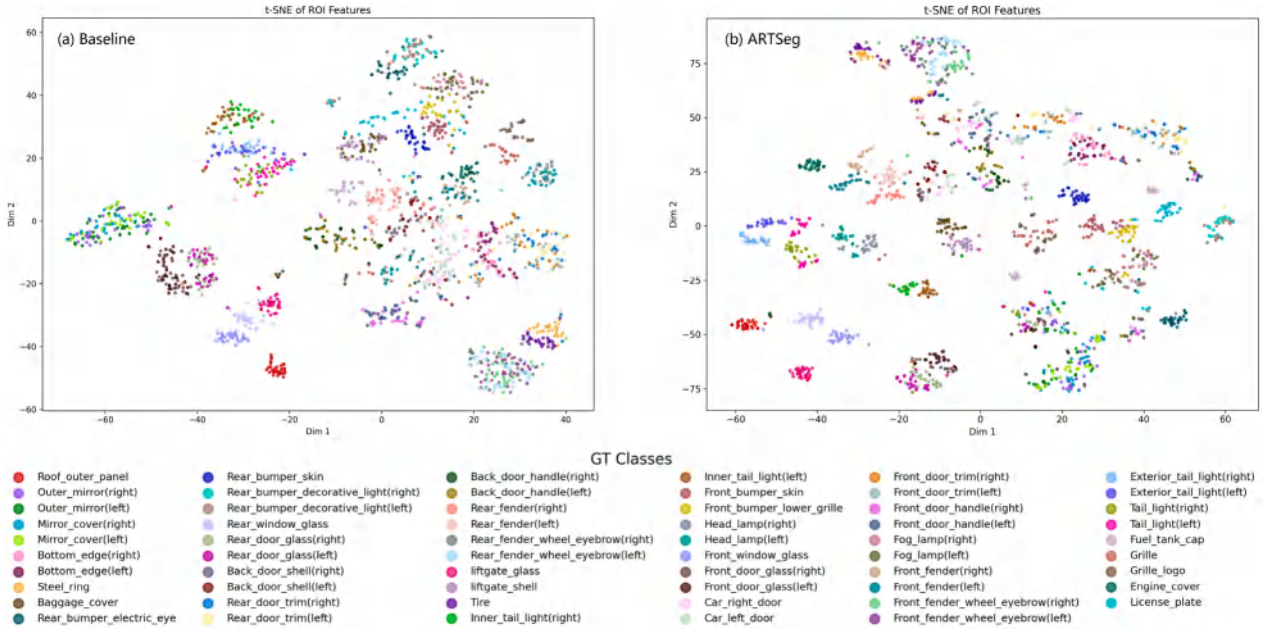


Fig. 12. The t-SNE-based feature visualization of intra-class clustering and inter-class separation on VCD dataset.

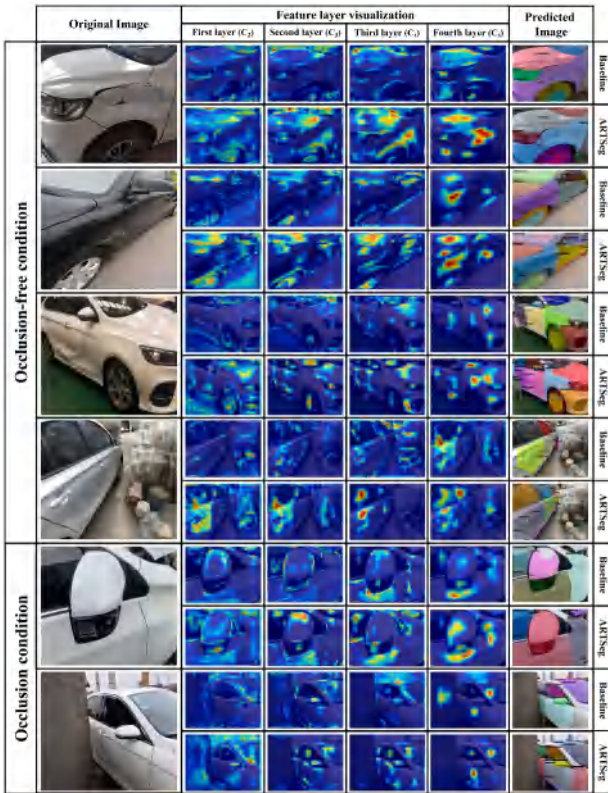


Fig. 13. Grad-CAM-based [49] backbone-level feature visualization on VCD dataset.

ture representations is ultimately reflected in the final prediction maps.

Furthermore, to qualitatively demonstrate the human-like feature attention strategy of ARTSeg in detecting both mirrored and non-mirrored vehicle component categories, we visualize the features upon which the model's categorical predictions are based, and compare them against the discriminative features of the baseline model, as shown in Fig. 14.

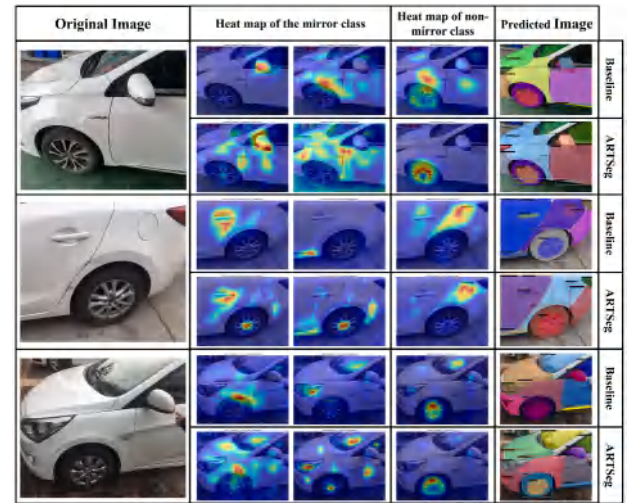


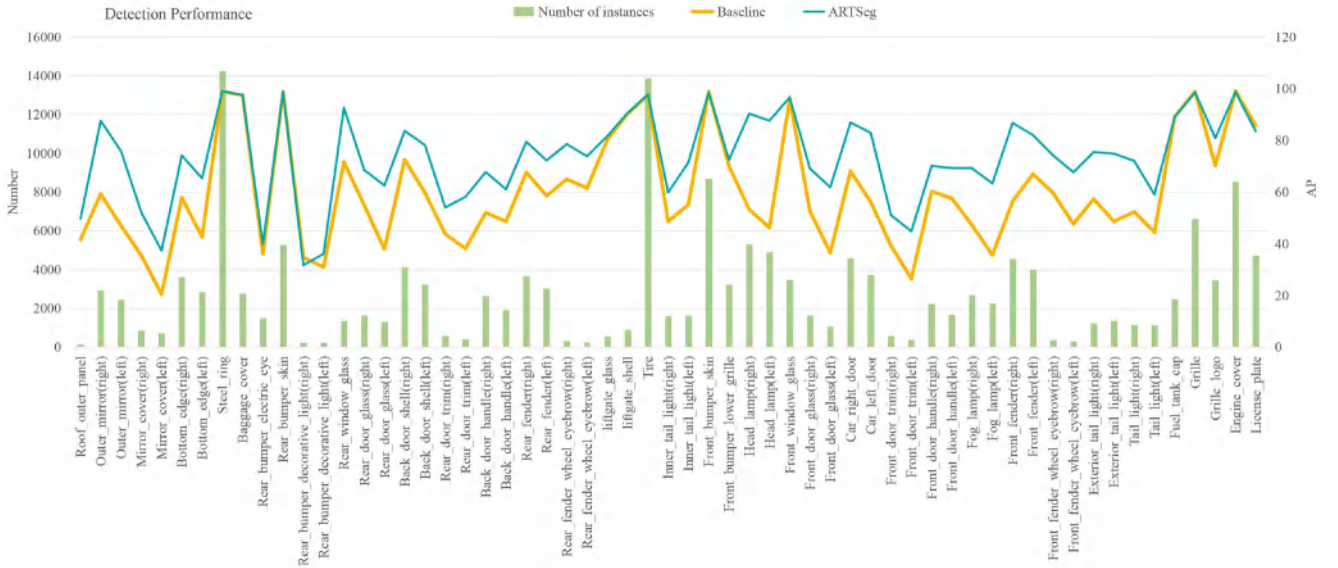
Fig. 14. Grad-CAM-based [49] feature visualization of saliency and discriminability for targets on VCD dataset.

For non-mirrored category recognition, the intra-class feature clustering constraint imposed by ICGC and the feature alignment introduced by RAFM during FPN fusion enhance the consistency of intra-class features, thereby strengthening the model's perception and attention on common and discriminative features of the component. This is reflected in the feature visualization, where the features extracted by ARTSeg for non-mirrored components are highly concentrated on the component itself and its surrounding area, and are more tightly clustered compared to the baseline model. This manner of reasoning is consistent with human cognition. For mirrored category recognition, ARTSeg models inter-class relationships via ICGC, enhancing the model's ability to capture latent structural correlations among vehicle component instances. This facilitates more effective handling of hidden topological structures in high-dimensional data. This is reflected in the feature visualization, where, compared to the baseline model, when identifying mirrored components, ARTSeg focuses not only on the component

Table 7

Detection and segmentation performance comparison across different component categories on VCD dataset.

Component category	Bbox_mAP ₅₀ (%)		Segm_mAP ₅₀ (%)		Component category	Bbox_mAP ₅₀ (%)		Segm_mAP ₅₀ (%)	
	Baseline	ARTSeg	Baseline	ARTSeg		Baseline	ARTSeg	Baseline	ARTSeg
Roof_outer_panel	41.6	49.6	41.6	50.2	Front_bumper_skin	98.9	98.5	98.7	98.3
Outer_mirror(right)	59.2	87.5	45.7	66.5	Front_bumper_lower_grille	70.0	72.4	70.0	72.2
Outer_mirror(left)	46.9	75.8	35.9	58.0	Rear_bumper_electric_eye	36.1	39.8	22.2	24.9
Mirror_cover(right)	35.5	52.1	37.6	52.0	Back_door_handle(right)	52.0	67.7	50.6	66.5
Mirror_cover(left)	20.4	37.3	21.6	36.7	Back_door_handle(left)	48.6	61.0	42.7	61.5
Bottom_edge(right)	57.8	74.1	52.5	74.6	Front_door_glass(right)	52.5	69.0	43.6	70.4
Bottom_edge(left)	42.6	65.3	38.3	64.0	Front_door_glass(left)	36.5	61.8	32.2	62.2
Steel_ring	98.9	99.0	98.9	99.0	Rear_fender_wheel_eyebrow(right)	64.9	78.5	67.1	79.6
Baggage_cover	97.4	97.5	97.4	97.7	Rear_fender_wheel_eyebrow(left)	61.5	73.8	66.2	73.8
Rear_bumper_skin	98.7	98.8	98.6	98.8	Front_door_trim(right)	39.2	51.0	36.6	51.2
Car_right_door	68.1	86.9	63.2	87.5	Front_door_trim(left)	26.4	44.8	25.5	45.8
Car_left_door	56.1	82.8	56.5	83.6	Front_door_handle(right)	60.2	70.1	57.8	69.6
Tail_light(right)	52.3	72.0	52.0	76.6	Front_door_handle(left)	57.5	69.2	54.7	68.5
Tail_light(left)	44.4	59.0	45.6	62.2	Rear_bumper_decorative_light(right)	34.7	31.7	27.7	33.8
Fog_lamp(right)	47.1	69.1	47.6	71.8	Rear_bumper_decorative_light(left)	30.9	36.1	30.8	37.1
Fog_lamp(left)	35.6	63.3	35.3	66.8	Rear_door_glass(right)	54.9	68.5	54.7	69.0
Front_fender(right)	56.4	86.7	53.9	87.6	Rear_door_glass(left)	37.9	62.5	34.8	63.0
Front_fender(left)	66.9	82.0	71.8	82.9	Front_fender_wheel_eyebrow(right)	59.6	74.1	57.4	74.1
Head_lamp(right)	53.2	90.3	53.6	92.0	Front_fender_wheel_eyebrow(left)	47.6	67.6	33.5	60.6
Head_lamp(left)	46.2	87.6	46.3	89.6	Exterior_tail_light(right)	57.3	75.4	54.8	81.4
Grille	98.9	98.5	98.7	98.4	Exterior_tail_light(left)	48.6	74.8	50.4	81.0
Grille_logo	70.2	80.9	70.1	81.0	Front_window_glass	95.9	96.6	96.0	96.6
Rear_fender(right)	67.6	79.4	69.4	80.5	Rear_window_glass	71.6	92.6	71.8	92.7
Rear_fender(left)	58.5	72.2	57.3	74.5	Inner_tail_light(right)	48.6	59.8	51.3	67.2
Engine_cover	99.1	98.7	99.2	98.7	Inner_tail_light(left)	55.1	71.4	52.6	77.6
License_plate	85.6	83.4	84.9	83.5	Back_door_shell(right)	72.5	83.6	63.3	84.5
Liftgate_glass	80.2	81.6	77.6	81.2	Back_door_shell(left)	59.7	78.1	46.9	79.2
Liftgate_shell	90.2	90.6	88.9	90.6	Rear_door_trim(right)	43.7	54.0	41.2	52.2
Tire	97.5	97.7	97.0	97.4	Rear_door_trim(left)	38.1	58.1	33.1	58.0
Fuel_tank_cap	89.1	89.1	89.0	89.2					

**Fig. 15.** Detection performance and quantity distribution across categories on VCD dataset.

itself and its surrounding region but also on image areas that assist in distinguishing mirrored orientations. This indicates that ARTSeg, in addition to considering the common features of the target component's category, also incorporates the vehicle's inherent structural characteristics and the spatial distribution among components within the image, in order to address the detection challenge of mirrored categories where components share identical intrinsic features but differ solely in spatial arrangement. Such a decision-making strategy is also consistent with human reasoning.

4.2.3. Comparative experiments

Table 7 compares the detection and segmentation accuracy across 59 component categories. Compared to the baseline model, ARTSeg achieves the best detection performance on 54 out of 59 categories, and the best segmentation performance on 55 categories. For the remaining components, although ARTSeg does not yield the highest accuracy, the performance gaps are within 2.2% for detection and 1.5% for segmentation compared to the best results.

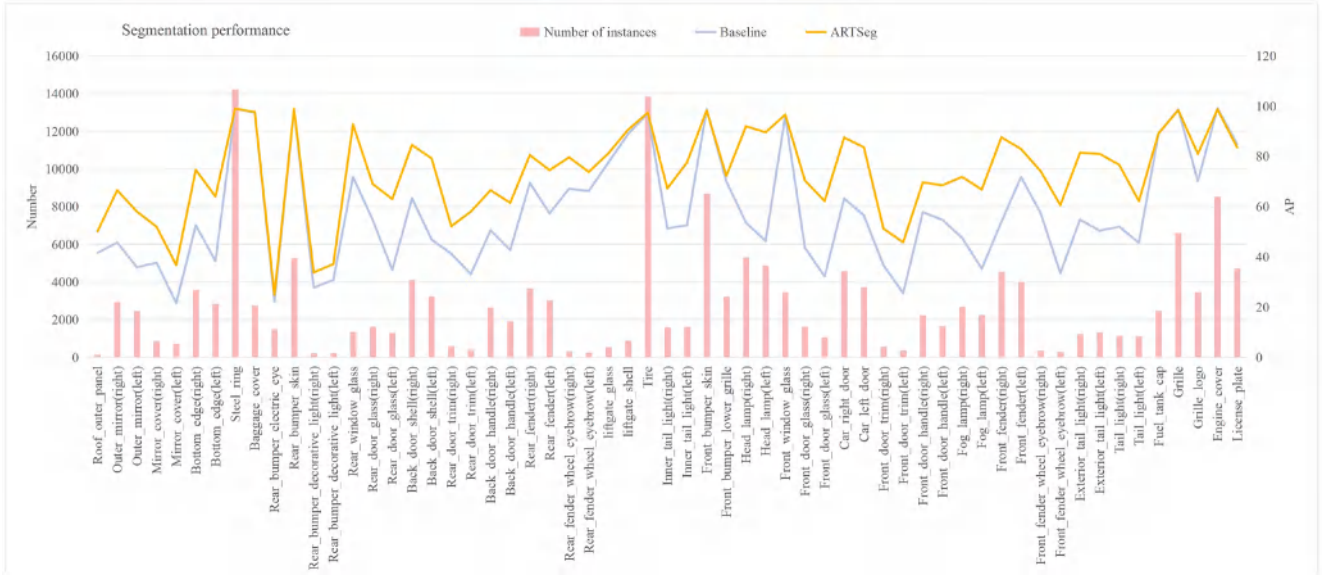


Fig. 16. Segmentation performance and quantity distribution across categories on VCD dataset.

Compared to the baseline model, ARTSeg leverages the ICGC to establish topological relationships among different categories, enabling it to distinguish between left-right mirrored components from a relational perspective. Additionally, it employs CDFB and TDS to optimize segmentation performance. As shown in Table 7, for the 42 mirrored component categories, ARTSeg improves both detection and segmentation performance on 41 of them, which is consistent with the conclusions drawn from the feature visualization experiments.

Moreover, some vehicle components appear infrequently in user-uploaded images for insurance assessment and are therefore underrepresented in the dataset, making them tail categories with lower detection and segmentation accuracy in the baseline model. ARTSeg enhances the representation of each category (particularly tail categories) within the feature space by synthesizing derived features through the ICGC. This facilitates the extraction of intra-class common features and promotes category-wise clustering, thereby improving the detection and segmentation performance of tail component categories to a certain extent. For example, the category *Front_fender_wheel_eyebrow* (left) has only 289 training images, accounting for just 1.5% of the 19,484 training samples in the VCD dataset. On the ARTSeg model, its detection and segmentation accuracy improved by 20.0% and 27.1%, respectively, compared to the baseline. To better illustrate the improvements ARTSeg brings to per-category detection and segmentation performance, Figs. 15 and 16 provide more intuitive visualizations.

To validate the superiority of our model, we conduct a comprehensive comparison with other instance segmentation algorithms on the Vehicle Component Dataset. The evaluation covers multiple aspects, including detection performance, segmentation accuracy, computational cost, and inference speed, as summarized in Table 8. Compared with other instance segmentation algorithms, the ARTSeg framework achieves a better balance between performance and computational cost, making it more aligned with practical application requirements.

We further conduct generalization experiments on two publicly available vehicle component datasets, Carparts [17] and DSMLR [18], with the results presented in Table 9. Compared with other instance segmentation algorithms, the ARTSeg model consistently demonstrates superior performance in recognizing vehicle components captured under varying application scenarios and conditions such as position and lighting.

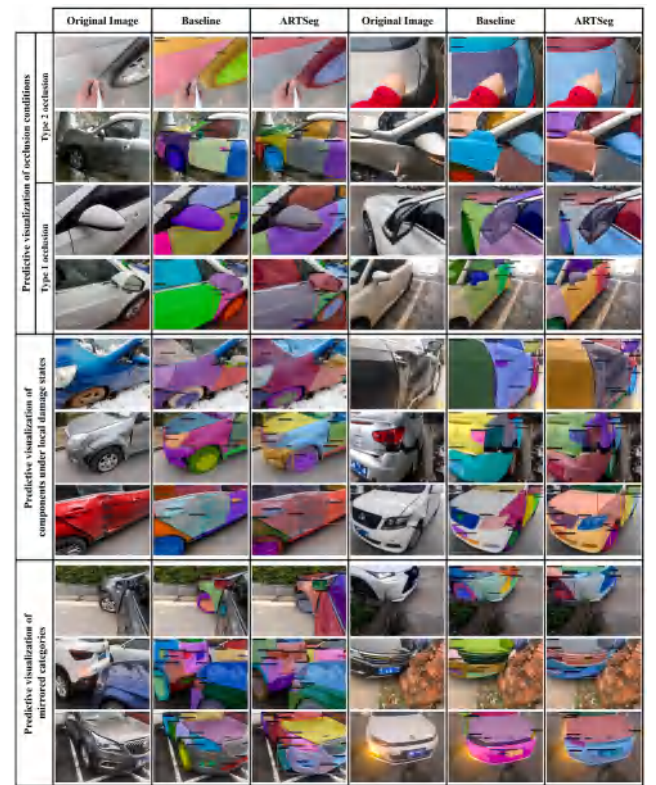


Fig. 17. Visualization of detection and segmentation predictions for 59 categories of vehicle components on VCD dataset.

4.2.4. Prediction visualization experiments

Based on complex component detection and segmentation scenarios in vehicle insurance damage assessment, we qualitatively compare the prediction visualizations between the baseline model and ARTSeg, as shown in Fig. 17. Compared to the baseline, ARTSeg demonstrates stronger adaptability to these challenging conditions. In occlusion sce-

Table 8
Comparative experiments with other models on VCD dataset.

Method	Backbone	Bbox		Segm		GPU	FPS	GFLOPs
		$AP_{50}(\%)$	$AP_{50-95}(\%)$	$AP_{50}(\%)$	$AP_{50-95}(\%)$			
(Baseline) Mask R-CNN [13]	ResNet-50	59.7	51.3	57.5	44.3	RTX3090Ti	31.89	183.4
Cascade Mask R-CNN [34]	ResNet-50	61.5	52.9	59.0	45.9	RTX3090Ti	22.91	275.6
DCT-Mask [32]	ResNet-50	62.5	53.5	61.3	48.1	RTX3090Ti	–	–
SparseInst [15]	ResNet-50	–	–	67.9	52.5	RTX3090Ti	40.63	83.9
Mask Transfuser [14]	ResNet-50	62.6	54.0	62.2	48.4	RTX3090Ti	12.11	659.8
Patch-DCT [30]	ResNet-50	63.7	54.7	63.1	48.7	RTX3090Ti	–	–
FastInst [16]	ResNet-50	–	–	70.1	53.5	RTX3090Ti	41.27	76.8
Yolov11 [29]	CSPDarkNet	71.1	61.2	68.8	53.0	RTX3090Ti	37.26	143.1
MTFDN [1]	ResNet-50	64.3	54.8	63.6	49.2	RTX3090Ti	25.13	221.6
(Textual Method) ARTSeg	ResNet-50	73.4	63.4	73.3	56.8	RTX3090Ti	20.56	302.8

Table 9
Comparative experiments on public datasets CarParts [17] and DSMLR [18].

Method	Carparts [17]				DSMLR [18]			
	Bbox		Segm		Bbox		Segm	
	$AP_{50}(\%)$	$AP_{50-95}(\%)$	$AP_{50}(\%)$	$AP_{50-95}(\%)$	$AP_{50}(\%)$	$AP_{50-95}(\%)$	$AP_{50}(\%)$	$AP_{50-95}(\%)$
(Baseline) Mask R-CNN [13]	66.5	52.4	66.1	52.0	59.1	39.5	60.1	41.5
Cascade Mask R-CNN [34]	68.6	54.7	67.5	53.5	61.9	44.7	61.2	43.5
DCT-Mask [32]	68.3	54.2	67.7	53.6	61.2	39.9	62.7	44.4
SparseInst [15]	–	–	68.3	53.9	–	–	63.1	42.6
Mask Transfuser [14]	68.7	54.0	68.5	53.7	61.5	39.9	62.4	43.2
Patch-DCT [30]	69.2	56.3	69.1	56.2	62.8	43.5	63.7	46.1
FastInst [16]	–	–	70.4	56.1	–	–	66.0	48.3
Yolov11 [29]	70.4	60.2	70.8	59.2	70.7	55.2	71.1	54.4
MTNDN [1]	69.8	56.9	69.6	56.7	64.0	45.0	64.3	46.5
(Textual Method) ARTSeg	73.4	62.1	73.0	62.0	72.6	58.6	72.8	57.3

Table 10
Generality evaluation of ARTSeg under various two-stage frameworks on VCD, CarParts [17], and DSMLR [18] datasets.

Algorithm framework	ARTSeg	VCD				Carparts [17]				DSMLR [18]			
		Bbox		Segm		Bbox		Segm		Bbox		Segm	
		$mAP_{50}(\%)$	$mAP_{50-90}(\%)$	$mAP_{50}(\%)$	$mAP_{50-90}(\%)$	$mAP_{50}(\%)$	$mAP_{50-90}(\%)$	$mAP_{50}(\%)$	$mAP_{50-90}(\%)$	$mAP_{50}(\%)$	$mAP_{50-90}(\%)$	$mAP_{50}(\%)$	$mAP_{50-90}(\%)$
Mask R-CNN [13]	–	59.7	51.3	57.5	44.3	66.5	52.4	66.1	52.0	59.1	39.5	60.1	41.5
	✓	73.4	63.4	73.3	56.8	73.4	62.1	73.0	62.0	72.6	58.6	72.8	57.3
MTFDN [1]	–	64.3	54.8	63.6	49.2	69.8	56.9	69.6	56.7	64.0	45.0	64.3	46.5
	✓	74.9	64.6	74.8	58.9	74.6	63.4	74.4	63.1	74.2	60.1	74.4	59.0

narios, ARTSeg effectively segments component boundaries even when vehicle components are partially occluded by hands or other components. In damage scenarios, ARTSeg maintains robust detection performance despite local deformation, misalignment, or breakage of components. For mirrored categories, ARTSeg accurately distinguishes between left and right components by leveraging vehicle structure and the spatial distribution of components. These results clearly demonstrate the practical advantages of ARTSeg in real-world applications.

4.2.5. Generality evaluation

It is worth noting that the algorithmic design of ARTSeg, which integrates inter-class relationship modeling, task-specific feature demand balancing, and general-purpose occlusion handling, is not restricted to Mask R-CNN [13]. Instead, it can be seamlessly incorporated into other two-stage instance segmentation frameworks, demonstrating strong generalization and compatibility. For instance, integrating ARTSeg into the MTFDN framework [1] leads to improved performance, as shown in Table 10.

5. Conclusion

In this paper, we propose an Adaptive Regularized Topological Segmentation network (ARTSeg) from the perspective of component detection and segmentation under the context of intelligent vehicle damage assessment. Within the ARTSeg framework, the Inter-Class Graph

Constraint (ICGC) module performs regularization on the backbone parameters, the Constrained Detail Feature Backtracking (CDFB) module adaptively adjusts feature resolutions for different instances, and the Topological Decoupling Segmentation (TDS) module conducts decoupled segmentation based on pixel-level relational modeling.

In the experimental validation stage, we conducted both quantitative and qualitative ablation studies on each proposed module and provided corresponding feature visualization comparisons. The results demonstrate that ARTSeg achieves significant improvements on the specific problems targeted by each module, validating their effectiveness and the synergistic interactions among them. Furthermore, comparative experiments with other models confirm the overall effectiveness of the proposed ARTSeg, which exhibits superior performance in both detection and segmentation tasks.

Extensive quantitative and qualitative experiments demonstrate that the proposed approach effectively addresses key challenges in component recognition tasks, including detection difficulties caused by significant intra-class variability and minimal inter-class differences, as well as segmentation challenges due to occlusions and unclear boundaries. By leveraging the three proposed modules, the method focuses on constructing inter-object relationships, balancing semantic and fine-grained detail information, and decoupling the modeling of occluded objects. As a result, it achieves superior performance in component detection and segmentation, offering an effective technical solution for recognizing multi-morphological components in complex scenarios.

In future work, we aim to integrate the proposed framework with damage assessment algorithms and deploy it in real-world vehicle intelligent assessment scenarios. From customer-uploaded damage images to the evaluation of vehicle component conditions, an end-to-end deployment and execution of intelligent vehicle damage assessment can be achieved.

CRedit authorship contribution statement

Xunqi Zhou: Writing – original draft, Software, Methodology; **Zhenqi Zhang:** Writing – review & editing, Validation, Funding acquisition; **Zifeng Wu:** Writing – review & editing, Validation; **Qianming Wang:** Supervision, Funding acquisition, Conceptualization; **Jing Teng:** Visualization, Investigation, Formal analysis; **Jinlong Liu:** Resources, Formal analysis, Data curation; **Yongjie Zhai:** Resources, Project administration, Funding acquisition.

Data availability

The data that has been used is confidential.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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